

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\173-C-CSChannel side-Final.pvdb

Date of Analysis : Sep 9,2026 9:42am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 173-C-_CS_Channel side-Final Page 2 of 200
 Vessel Design Summary: Step: 32 9:43am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1050.000 mm ID
 Vessel Design Length, Tangent to Tangent 8874.95 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 275 °C
 Channel Side Design Temperature 82 °C
 Shell Side Design Pressure 1080.000 KPa.
 Channel Side Design Pressure 1000.000 KPa.
 Shop Shell Side Test Pressure 2082.111 KPa.
 Field Shell Side Test Pressure 1404.000 KPa.
 Shop Channel Side Test Pressure 2082.111 KPa.
 Field Channel Side Test Pressure 1300.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Flange	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	1010.321	103.42	1436.000	3.0000	No	No
LHS Ch Shell	1010.321	103.42	2829.000	3.0000	N/A	No
LHS Ch Flg	1010.488	103.42	2520.000	3.0000	N/A	No
LHS Shell Flg	1090.488	0.00	2363.000	3.0000	N/A	No
Main Shell	1090.322	0.00	2808.000	3.0000	N/A	No
RHS Dish End	1090.322	0.00	1426.000	3.0000	No	No
Body Flg	1080.000	0.00	3139.000	3.0000	N/A	No

Liquid Level: 1070.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.8	50.8	14.0	8.5	6.8	5.8	1.00	1.00
Cylinder	750.8	700.0	14.0	14.0	6.9	6.3	1.00	1.00
Body Flg	897.1	140.0	115.0	107.0	103.6	103.6	1.00	1.00
Body Flg	1124.1	140.0	115.0	107.0	103.6	103.6	1.00	1.00
Cylinder	8824.2	7700.0	14.0	14.0	7.2	...	1.00	1.00
Ellipse	8875.0	50.8	14.0	8.5	7.2	...	1.00	1.00
Body Flg	4830.8	0.0	52.0	44.0	41.3	41.3	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.957	Nozzle	Passed
N2-Tube Outlet	WRC-297	0.957	Nozzle	Passed
N3-Shell Inlet	WRC-297	0.770	Nozzle	Passed
N4-Shell Inlet	WRC-297	0.770	Nozzle	Passed
N5-Shell Outle	WRC-297	0.686	Shell	Passed

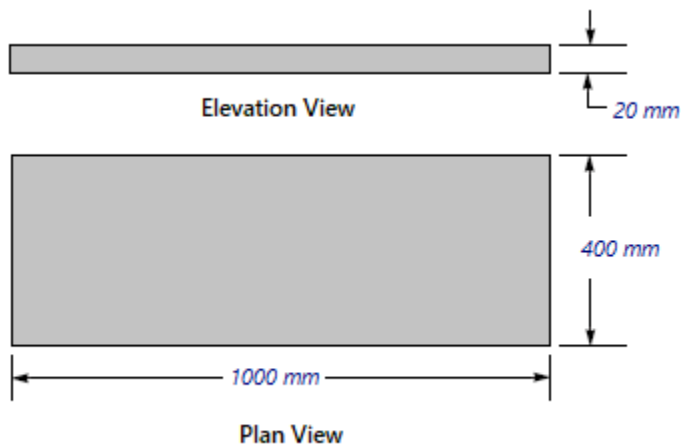
Foundation/Support Summary

Saddle Parameters:

Saddle Width	375.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1000.000 mm
Wear Pad Width	510.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1540.000 mm

Baseplate Length	1000.000 mm
Baseplate Thickness	20.000 mm
Baseplate Width	400.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	475.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch





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Vessel Design Summary: Step: 32 9:43am Sep 9,2026

Maximum Tensile Bolt Load 1596. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	16426.78	Kgf
Maximum Transverse Saddle Shear Load	618.27	Kgf
Maximum Longitudinal Saddle Shear Load	1236.54	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	14092.99	Kgf
Maximum Transverse Saddle Shear Load	131.50	Kgf
Maximum Longitudinal Saddle Shear Load	56.47	Kgf

Weights:

Fabricated - Bare w/o removable internals	12561.9	kg.
Shop Test - Fabricated + water (full)	20410.1	kg.
Shipping - Fab. + rem. intls.+ shipping app.	12561.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	13061.9	kg.
Empty - Fab. + intls. + details + wghts.	14171.7	kg.
Operating - Empty + operating liquid (no ca)	21498.8	kg.
Field Test - Empty weight + test medium (full)	20902.3	kg.

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 FileName : 173-C-_CS_Channel side-Final Page 5 of 200
 Nozzle Summary: Step: 30 9:43am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	1562	OK	...	OK	6.84	OK	Passed
N2-Tube Outlet	1562	OK	...	OK	6.88	OK	Passed
N3-Shell Inlet	2808	...	...	OK	7.18	OK	Passed
N4-Shell Inlet	2808	...	...	OK	7.18	OK	Passed
N5-Shell Outlet	2808	...	...	OK	7.22	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2808 Nozzle : N3-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 1562 Nozzle : N1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1-Tube Inlet	400.800	0.000	408.900
20	N2-Tube Outlet	400.800	180.000	408.900
50	N3-Shell Inlet	3024.150	0.000	602.601
50	N4-Shell Inlet	6824.150	0.000	602.601
50	N5-Shell Outlet	4924.150	180.000	850.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	8.000 in	60	WNF	8.625	10.312	320.00	14.00	197.55	150
N2-Tube Outlet	8.000 in	60	WNF	8.625	10.312	320.00	14.00	197.55	150
N3-Shell Inlet	12.000 in	60	WNF	12.750	14.275	473.85	14.00	243.60	300
N4-Shell Inlet	12.000 in	60	WNF	12.750	14.275	473.85	14.00	243.60	300
N5-Shell Outlet	18.000 in	60	WNF	18.000	19.050	607.20	14.00	274.38	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-106 B	14.000	10.000	12.000	14.000	...
N2-Tube Out	SA-106 B	14.000	10.000	12.000	14.000	...
N3-Shell In	SA-106 B	14.000	10.000	12.000	14.000	...
N4-Shell In	SA-106 B	14.000	10.000	12.000	14.000	...
N5-Shell Ou	SA-106 B	14.000	10.000	12.000	14.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	400.800	0.0	170.00	0.00	LHS Ch Shell
N2-Tube Outlet	400.800	180.0	170.00	0.00	LHS Ch Shell
N3-Shell Inlet	3024.150	0.0	200.00	0.00	Main Shell
N4-Shell Inlet	6824.150	0.0	200.00	0.00	Main Shell
N5-Shell Outlet	4924.150	180.0	200.00	0.00	Main Shell

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Bill of Material: Step: 28 9:43am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 1050.0mm ID X 50.8mm	SA-516 70
1	CYLINDER: 14.0mm THK X 1050.0mm ID X 700.0mm	SA-516 70
2	WELD NECK FLANGE: 115.0mm THK X 1230.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 1050.0mm ID X 7700.0mm	SA-516 70
1	WELD NECK FLANGE: 52.0mm THK X 635.0mm OD	SA-105
10	BAFFLES: 1044.0mm DIA X 8.0mm THK	SA-516 70
8	TIE RODS: 14mm DIA X 6006mm	SA-193 B7M
1	TUBESHEET: 1230.0mm X 87.0mm THK	SA-266 2
668	TUBES: 7350.0mm X 25.4mm DIA X 1.6mm THK	SA-213 TP304L
1	INSULATION: 104mm X 50mm THK	Hot Insulation
2	SADDLE: 375.0mm X 120 DEG	SA-516 70
1	INSULATION: 7700mm X 100mm THK	Hot Insulation
1	INSULATION: 50mm X 100mm THK	Hot Insulation
1	GASKET: 1114.0mm OD X 1082.0mm ID	...
60	BODY FLANGE BOLTS: 28.6mm DIA	SA-193 B7
104	NUTS FOR BODY FLANGE BOLTS: 28.6mm DIA	...
1	GASKET: 527.0mm OD X 474.7mm ID	...
1	NAMEPLATE	...



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 MDMT Summary: Step: 31 9:43am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
LHS Shell Flg[11]	B	-18	-48	-29	0.469	14.000	1.00	No
[11]	A	11	-104		0.167	19.050	1.00	No
Bolting [21]		-48						
Main Shell [8]	D	-48	-48	-29	0.499	14.000	1.00	No
RHS Dish End [10]	B	-29	-30	-29	0.986	8.500	1.00	No
RHS Dish End [7]	B	-18	-48	-29	0.499	14.000	1.00	No
Left Saddle F[24]	D	-48	-48	-29	0.499	10.000	1.00	No
RH Fixed sadd[24]	D	-48	-48	-29	0.499	10.000	1.00	No
N3-Shell Inlet[1]	B	-1	-104	-29	0.156	12.490	1.00	No
Nozzle Flg [5]	A	-18	-104					
N4-Shell Inlet[1]	B	-1	-104	-29	0.156	12.490	1.00	No
Nozzle Flg [5]	A	-18	-104					
N5-Shell Outle[1]	B	7	-104		0.154	16.669	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: SS[13]	B	-5	-23	-29	0.671	21.750	1.00	No
Warmest MDMT:		11	-23					
LHS Ch Flg [11]	B	-18	-48	-29	0.352	14.000	1.00	No
LHS Dish End [10]	B	-29	-43	-29	0.739	8.500	1.00	No
LHS Dish End [7]	B	-18	-48	-29	0.374	14.000	1.00	No
LHS Ch Shell [8]	D	-48	-48	-29	0.374	14.000	1.00	No
N1-Tube Inlet [1]	B	-8	-104	-29	0.154	9.023	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2-Tube Outlet[1]	B	-8	-104	-29	0.155	9.023	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tubesheet: CS[14]	B	-5	-23	-29	0.671	21.750	1.00	No
Bolting [21]		-48						
Warmest MDMT:		-5	-23					
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	-29.0		-12.0		Pass			
Channel/Tube	-29.0		-12.0		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material



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MDMT Summary:

Step: 31

9:43am

Sep 9, 2026

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors: Step: 0 9:42am Sep 9,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: LHS Ch Flg and LHS Shell Flg

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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 Input Echo: Step: 1 9:42am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1080	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	275.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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 Input Echo: Step: 1 9:42am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	8050 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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 FileName : 173-C-_CS_Channel side-Final Page 13 of 200
 Input Echo: Step: 1 9:42am Sep 9,2026

Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	LHS Dish End	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	1050	mm
Element Thickness	8.5	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1050	mm
Liquid Density	0.0009984	kg./cm ³



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Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	700 mm
Inside Diameter	1050 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1050 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	350 mm
Nozzle Diameter	8 in.
Nozzle Schedule	60
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	33.315 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	20
Detail Type	Nozzle
Detail ID	N2-Tube Outlet
Dist. from "FROM" Node / Offset dist	350 mm
Nozzle Diameter	8 in.
Nozzle Schedule	60
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	33.315 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	20
Detail Type	Weight
Detail ID	Partition weigh
Dist. from "FROM" Node / Offset dist	350 mm
Miscellaneous Weight	300 Kgf
Offset from Element Centerline	0 mm

Element# 3/7
 Element From Node 30



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Element To Node	40
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	140 mm
Flange Inside Diameter	1050 mm
Element Thickness	107 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	115 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1070 mm
Liquid Density	0.0009984 kg./cm ³

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Flange
Description	LHS Shell Flg
Distance "FROM" to "TO"	140 mm
Flange Inside Diameter	1050 mm
Element Thickness	107 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	115 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1080 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	275 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1070 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	40



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Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	104 mm
Thickness of Insulation	50 mm
Density	0.000352 kg./cm ³

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	7700 mm
Inside Diameter	1050 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1080 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	275 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	137 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	210 N./mm ²
UCS-66 Chart Curve Designation	D
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle Flo
Dist. from "FROM" Node / Offset dist	1540 mm
Width of Saddle	375 mm
Height of Saddle at Bottom	1000 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	510 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.3000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	50
Detail Type	Saddle
Detail ID	RH Fixed saddle
Dist. from "FROM" Node / Offset dist	6160 mm
Width of Saddle	375 mm
Height of Saddle at Bottom	1000 mm
Saddle Contact Angle	120.0



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Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	510	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1050	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	7700	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	1900	mm
Nozzle Diameter	12	in.
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.01	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N4-Shell Inlet	
Dist. from "FROM" Node / Offset dist	5700	mm
Nozzle Diameter	12	in.
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.01	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N5-Shell Outlet	
Dist. from "FROM" Node / Offset dist	3800	mm
Nozzle Diameter	18	in.
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	205.72	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



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Element From Node	50
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	1540 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1000 mm

Element From Node	50
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	6160 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1000 mm

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	RHS Dish End
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	1050 mm
Element Thickness	8.5 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1080 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	275 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1050 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50.5 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	
Distance "FROM" to "TO"	144.08 mm
Flange Inside Diameter	418.9 mm
Element Thickness	44 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	52 mm



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External Corrosion Allowance	0	mm
Design Internal Pressure	1080	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-105	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	132.5	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	199	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K03504	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

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XY Coordinate Calculations: Step: 2 9:42am Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Dish End		50.8	...	50.8	...
LHS Ch Shell		750.8	...	700	...
LHS Ch Flg		897.15	...	140	...
LHS Shell Flg		1124.15	...	140	...
Main Shell		8824.15	...	7700	...
RHS Dish End		8874.95	...	50.8	...
70	80	4830.8	...	...	144.076

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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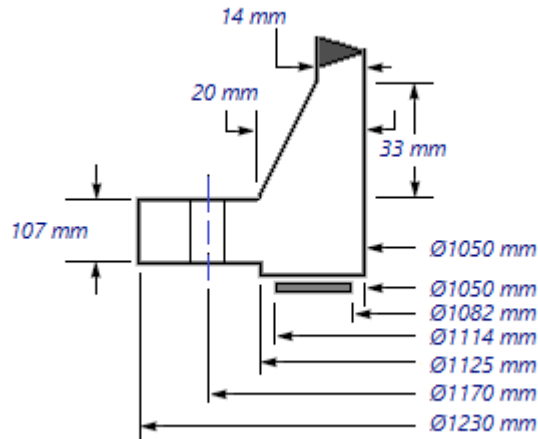
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 Flg Calc [Int P]: New Flange Flng: 16 9:42am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1010.49	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1050.000	mm
Flange Outside Diameter	A	1230.000	mm
Flange Thickness	t	107.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	33.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1170.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		44	
Flange Face Outside Diameter	Fod	1125.000	mm
Flange Face Inside Diameter	Fid	1050.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1114.000	mm
Gasket Inside Diameter	Gi	1082.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1102.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²
Mating Flange Operating Bolt Load		129912.99	Kgf
Mating Flange Seating Bolt Load		215021.02	Kgf
Mating Flange Design Bolt Load		362446.06	Kgf



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1010 \cdot (1050/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1010) + Ca$$

$$= 6.8835 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11)$$

$$= 2839.341 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (525 + 0.6 \cdot 14)$$

$$= 3621.837 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	104.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1056.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	40.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $b_o = N/2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1099.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1100^2 \cdot 1010$$

$$= 97881.758 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1100 \cdot 3 \cdot 1010$$

$$+ 2 \cdot 1102 \cdot 5 \cdot 3 \cdot 1010$$

$$= 18631.512 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1056^2 \cdot 1010$$

$$= 90249.656 \text{ Kgf}$$



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 Flg Calc [Int P]: New Flange Flng: 16 9:42am Sep 9,2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 97882 - 90250$$

$$= 7632.104 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(97882 + 18632 + 0, 0)$$

$$= 116513.273 \text{ Kgf}$$

$$= 129912.992 \text{ Kgf, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1100 + 68.9 \cdot 5 \cdot 1102$$

$$= 211872.547 \text{ Kgf}$$

$$= 215021.016 \text{ Kgf, Mating Flange Load Governs}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((129913 + 0 + 4 \cdot 0 / 1100) / 172, 215021 / 172)$$

$$= 122.600 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 104 / (3 + 0.5)$$

$$= 235.436 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1170 \cdot \sin(\pi / 44)$$

$$= 83.467 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 83.5 / (2 \cdot 28.6 + 104) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	122.600	206.658	
Radial Distance between Hub and Bolts:	38.100	40.000	
Radial Distance between Bolts and Edge:	28.575	30.000	
Circ. Spacing between the Bolts:	63.500	83.467	235.436

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 207 \cdot 172 / (68.9 \cdot \pi (1114 + 1082))$$

$$= 7.472 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 207$$

$$= 362446.06 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 129913 - 97882$$



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 Flg Calc [Int P]: New Flange Flng: 16 9:42am Sep 9,2026

= 32031.23 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1170 - 1100) / 2$
 $= 35.1274 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((1170 - 1056) / 2 + 35.1)$
 $= 46.0637 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (1170 - 1056 - 17) / 2$
 $= 48.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	90250.	48.5000	1.0000	4377.
Face Pressure, Mt	7632.	46.0637	1.0000	352.
Gasket Load, Mg	32031.	35.1274	1.0000	1125.
Gasket Seating, Matm	362446.	35.1274	1.0000	12732.

Total Moment for Operation, Mo 5854. Kg-m.
 Total Moment for Gasket seating, Mg 12732. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 107.778 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.306
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.545

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.879
 Factor V per 4.16.5 0.360
 Factor f per 4.16.5 1.224

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 1230/1056$
 $= 1.165$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.16^2 \cdot \log_{10}(1.16))) / (1.16^2 - 1)) / (1.16 - 1)$
 $= 12.798$

Stress factor from Table 4.16.4 [T]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.16^2(1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / ((1.0472 + 1.9448 \cdot 1.16^2) \cdot (1.16 - 1))$
 $= 1.853$

Stress factor from Table 4.16.4 [U]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.16^2(1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / (1.36136 (1.16^2 - 1) \cdot (1.16 - 1))$
 $= 14.064$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.16^2 + 1) / (1.16^2 - 1)$
 $= 6.607$



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Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 14.1 \cdot 11^2 \cdot 108 / 0.36$$

$$= 509961.875 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88 / 108$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (104 \cdot 0.0082 + 1) / 1.85 + 104^3 / 509962$$

$$= 3.203$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.22 \cdot 5854 / 1056) / (3.2 \cdot 17^2)$$

$$= 71.89 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.22 \cdot 12732 / 1056) / (3.2 \cdot 17^2)$$

$$= 156.36 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.13 \cdot 5854 / 1056) / (3.2 \cdot 104^2)$$

$$= 3.34 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.13 \cdot 12732 / 1056) / (3.2 \cdot 104^2)$$

$$= 7.27 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (12.8 \cdot 5854 / (104^2 \cdot 1056)) - 6.61 \cdot 3.34$$

$$= 42.24 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (12.8 \cdot 12732 / (104^2 \cdot 1056)) - 6.61 \cdot 7.27$$

$$= 91.86 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (71.9 + \max(3.34, 42.2)) / 2$$

$$= 57.06 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (156 + \max(7.27, 91.9)) / 2$$

$$= 124.11 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 129913 / 207$$

$$= 61.65 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (215021 / 207)$$



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= 102.04 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	71.89	207.00	156.36	207.00
Radial Flange	3.34	138.00	7.27	138.00
Tangential Flange	42.24	138.00	91.86	138.00
Maximum Average	57.06	138.00	124.11	138.00
Bolting	61.65	172.00	102.04	172.00

Minimum Required Flange Thickness [Rigidity] 103.556 mm
 Estimated M.A.W.P. (Operating) 2531 KPa.
 Estimated Finished Weight of Flange at given Thk. 281.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 349.7 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	295.13	374.99	0.787	295.13	374.99	0.787
Tangential Flange	Sto	143.27	250.00	0.573	143.27	250.00	0.573
Combined	Sh/f + St	705.49	749.99	0.941	705.49	749.99	0.941
Combined	Sr + St	318.49	749.99	0.425	318.49	749.99	0.425
Combined	Sh + Sto	643.27	749.99	0.858	643.27	749.99	0.858
Radial Flange	Sr	23.36	499.99	0.047	23.36	499.99	0.047
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.53/1.25)
 = 1.2184

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.42/1.19) if h/ho < 0.35, else max (0.25; 0.28/0.77)
 = 0.3491

In this case, the hub ratio h/ho, 0.3062 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	552.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.491 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (1114^2 - 1082^2) + (0.5 \cdot 1102 \cdot 5)$
 $= 579.5 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207 \cdot 579 / (44 \cdot 4.7)$
 $= 580.4 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(580; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 553)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (579 / (44 \cdot 4.7))$
 $:: 507.5 \geq 392.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 579 + \pi/4 \cdot 1010 \cdot 1082^2) / (0.7 \cdot 4.7 \cdot 44)$
 $:: 507.5 \geq 452.8 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (579 / (44 \cdot 4.7))$
 $:: 507.5 \leq 773.9 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 508 \leq 553 (1/0.49)$
 $:: 507.5 \leq 1125.8 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 4.7 \cdot 28.6$
 $= 138.9 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12732 / 1 \cdot 9807 \cdot 0.36 / (2.98 \cdot 201334 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5854 / 1 \cdot 9807 \cdot 0.36 / (2.98 \cdot 197847 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.467$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12732 / 1 \cdot 9807 \cdot 0.36 / (3.2 \cdot 201334 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.928$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5854 / 1 \cdot 9807 \cdot 0.36 / (3.2 \cdot 197847 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.434$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.35, Temperature Reduction per Fig. UCS 66.1 = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	64.99	207.00	137.95	207.00
Radial Flange	4.06	138.00	8.62	138.00
Tangential Flange	34.52	138.00	73.28	138.00
Maximum Average	49.76	138.00	105.61	138.00
Bolting	61.65	172.00	102.04	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 90.068 mm

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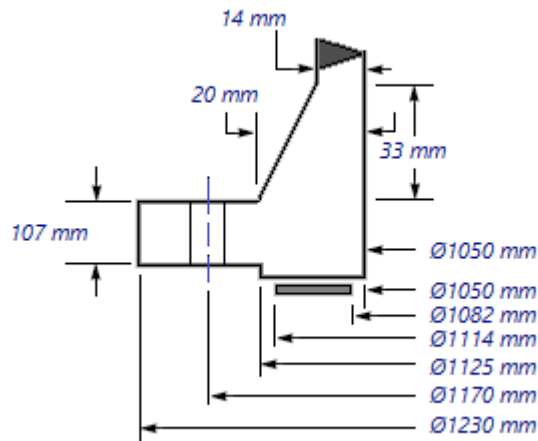
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Flange Input Data Values

Description: LHS Shell Fl :

LHS Shell Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1090.49	KPa.
Design Temperature		275	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1050.000	mm
Flange Outside Diameter	A	1230.000	mm
Flange Thickness	t	107.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	33.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	132.50	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1170.000	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		44	
Flange Face Outside Diameter	Fod	1125.000	mm
Flange Face Inside Diameter	Fid	1050.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1114.000	mm
Gasket Inside Diameter	Gi	1082.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1102.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1090 \cdot (1050/2 + 3)) / (137 \cdot 1 - 0.6 \cdot 1090) + C_a$$

$$= 7.2232 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11)$$

$$= 2818.767 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (525 + 0.6 \cdot 14)$$

$$= 3621.837 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	104.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1056.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	17.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	40.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $b_o = N/2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1099.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1100^2 \cdot 1090$$

$$= 105631.031 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1100 \cdot 3 \cdot 1090$$

$$+ 2 \cdot 1102 \cdot 5 \cdot 3 \cdot 1090$$

$$= 20106.566 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1056^2 \cdot 1090$$

$$= 97394.703 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 105631 - 97395$$

$$= 8236.334 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(105631 + 20107 + 0, 0)$$

$$= 125737.602 \text{ Kgf}$$

$$= 129912.992 \text{ Kgf, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1100 + 68.9 \cdot 5 \cdot 1102$$

$$= 211872.547 \text{ Kgf}$$

$$= 215021.016 \text{ Kgf, Mating Flange Load Governs}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((129913 + 0 + 4 \cdot 0 / 1100) / 172, 215021 / 172)$$

$$= 122.600 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 104 / (3 + 0.5)$$

$$= 235.436 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1170 \cdot \sin(\pi / 44)$$

$$= 83.467 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 83.5 / (2 \cdot 28.6 + 104) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	122.600	206.658	
Radial Distance between Hub and Bolts:	38.100	40.000	
Radial Distance between Bolts and Edge:	28.575	30.000	
Circ. Spacing between the Bolts:	63.500	83.467	235.436

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 207 \cdot 172 / (68.9 \cdot \pi (1114 + 1082))$$

$$= 7.472 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 207$$

$$= 362446.06 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 129913 - 105631$$



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= 24281.96 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1170 - 1100) / 2$
 $= 35.1274 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((1170 - 1056) / 2 + 35.1)$
 $= 46.0637 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (1170 - 1056 - 17) / 2$
 $= 48.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	97395.	48.5000	1.0000	4724.
Face Pressure, Mt	8236.	46.0637	1.0000	379.
Gasket Load, Mg	24282.	35.1274	1.0000	853.
Gasket Seating, Matm	362446.	35.1274	1.0000	12732.

Total Moment for Operation, Mo 5956. Kg-m.
 Total Moment for Gasket seating, Mg 12732. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 107.778 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.306
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.545

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.879
 Factor V per 4.16.5 0.360
 Factor f per 4.16.5 1.224

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 1230/1056$
 $= 1.165$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.16^2 \cdot \log_{10}(1.16))) / (1.16^2 - 1)) / (1.16 - 1)$
 $= 12.798$

Stress factor from Table 4.16.4 [T]:
 $= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.16^2(1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / ((1.0472 + 1.9448 \cdot 1.16^2) \cdot (1.16 - 1))$
 $= 1.853$

Stress factor from Table 4.16.4 [U]:
 $= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 $= (1.16^2(1 + 8.55246 \cdot \log_{10}(1.16)) - 1) / (1.36136(1.16^2 - 1) \cdot (1.16 - 1))$
 $= 14.064$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$
 $= (1.16^2 + 1) / (1.16^2 - 1)$
 $= 6.607$



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Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 14.1 \cdot 11^2 \cdot 108 / 0.36$$

$$= 509961.875 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88 / 108$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (104 \cdot 0.0082 + 1) / 1.85 + 104^3 / 509962$$

$$= 3.203$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.22 \cdot 5956 / 1056) / (3.2 \cdot 17^2)$$

$$= 73.15 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.22 \cdot 12732 / 1056) / (3.2 \cdot 17^2)$$

$$= 156.36 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.13 \cdot 5956 / 1056) / (3.2 \cdot 104^2)$$

$$= 3.40 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.13 \cdot 12732 / 1056) / (3.2 \cdot 104^2)$$

$$= 7.27 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (12.8 \cdot 5956 / (104^2 \cdot 1056)) - 6.61 \cdot 3.4$$

$$= 42.97 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (12.8 \cdot 12732 / (104^2 \cdot 1056)) - 6.61 \cdot 7.27$$

$$= 91.86 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (73.1 + \max(3.4, 43)) / 2$$

$$= 58.06 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (156 + \max(7.27, 91.9)) / 2$$

$$= 124.11 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm_1 / Ab$$

$$= 129913 / 207$$

$$= 61.65 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (215021 / 207)$$



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= 102.04 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	73.15	198.75	156.36	207.00
Radial Flange	3.40	132.50	7.27	138.00
Tangential Flange	42.97	132.50	91.86	138.00
Maximum Average	58.06	132.50	124.11	138.00
Bolting	61.65	172.00	102.04	172.00

Minimum Required Flange Thickness [Rigidity] 103.556 mm

Estimated M.A.W.P. (Operating) 2373 KPa.
 Estimated Finished Weight of Flange at given Thk. 281.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 349.7 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	454.86	454.86	1.000	454.86	499.99	0.910
Tangential Flange	St	268.49	341.14	0.787	268.49	374.99	0.716
Tangential Flange	Sto	130.34	227.43	0.573	130.34	250.00	0.521
Combined	Sh/f + St	641.80	682.29	0.941	641.80	749.99	0.856
Combined	Sr + St	289.75	682.29	0.425	289.75	749.99	0.386
Combined	Sh + Sto	585.20	682.29	0.858	585.20	749.99	0.780
Radial Flange	Sr	21.25	454.86	0.047	21.25	499.99	0.043
Maximum Stress Ratio				1.000			0.910



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den})$$

$$= \max(1; 1.53/1.25)$$

$$= 1.2184$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3})$$

$$= \max(0.25; 0.42/1.19) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.28/0.77)$$

$$= 0.3491$$

In this case, the hub ratio $h/h0$, 0.3062 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	502.7 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	276.0 N./mm ²
Minimum gasket operating stress	Sg_{min-O}	97.0 N./mm ²
Minimum gasket seating stress	Sg_{min-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700



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Root Area of a Single Bolt	Abs	4.6968 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	199.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.447 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (1114^2 - 1082^2) + (0.5 \cdot 1102 \cdot 5)$
 $= 579.5 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207 \cdot 579 / (44 \cdot 4.7)$
 $= 580.4 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(580; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 503)$
 $= 502.7 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 503 \geq 140 (579 / (44 \cdot 4.7))$
 $:: 502.7 \geq 392.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 503 \geq (97 \cdot 579 + \pi/4 \cdot 1090 \cdot 1082^2) / (0.7 \cdot 4.7 \cdot 44)$
 $:: 502.7 \geq 457.9 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 503 \leq 276 (579 / (44 \cdot 4.7))$
 $:: 502.7 \leq 773.9 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 503 \leq 503 (1/0.45)$
 $:: 502.7 \leq 1125.8 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 503 \cdot 0.2 \cdot 4.7 \cdot 28.6$
 $= 137.6 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	112 N./mm ²
Bolt Allowable Stress for: SA-193 B7	Sb	171 N./mm ²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12732 / 1 \cdot 9807 \cdot 0.36 / (2.98 \cdot 201334 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5956 / 1 \cdot 9807 \cdot 0.36 / (2.98 \cdot 185553 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.507$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12732 / 1 \cdot 9807 \cdot 0.36 / (3.2 \cdot 201334 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.928$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 5956 / 1 \cdot 9807 \cdot 0.36 / (3.2 \cdot 185553 \cdot 11^2 \cdot 108 \cdot 0.3)$
 $= 0.471$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.47, Temperature Reduction per Fig. UCS 66.1 = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	66.22	207.00	137.95	207.00
Radial Flange	4.14	138.00	8.62	138.00
Tangential Flange	35.18	138.00	73.28	138.00
Maximum Average	50.70	138.00	105.61	138.00
Bolting	61.65	172.00	102.04	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 90.068 mm

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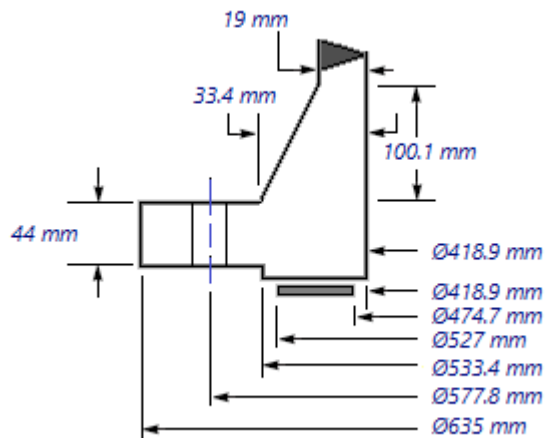
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Flange Input Data Values

Description: FLANGE :

Item: Node 70 to 80

Flange Type	Integral Weld Neck		
Design Pressure	P	1080.00	KPa.
Design Temperature		275	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	418.900	mm
Flange Outside Diameter	A	635.000	mm
Flange Thickness	t	44.0000	mm
Thickness of Hub at Small End	go	19.0500	mm
Thickness of Hub at Large End	gl	33.4010	mm
Length of Hub	h	100.0760	mm
Flange Material		SA-105	
Flange Material UNS number		K03504	
Flange Allowable Stress At Temperature	Sfo	132.50	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	577.850	mm
Nominal Bolt Diameter	a	28.5750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	533.400	mm
Flange Face Inside Diameter	Fid	418.900	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	527.050	mm
Gasket Inside Diameter	Gi	474.726	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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The rigidity index calculation has been turned off. No exceptions are allowed in VIII-2 part 4.16. The following results are for information only.

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1080 \cdot (419/2 + 3)) / (133 \cdot 1 - 0.6 \cdot 1080) + Ca$$

$$= 4.7403 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133 \cdot 1 \cdot 16) / (212 + 0.6 \cdot 16)$$

$$= 9575.448 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 19) / (209 + 0.6 \cdot 19)$$

$$= 11901.031 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	41.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	424.900 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.401 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	16.050 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	46.074 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	26.162 mm
Basic Gasket Width, $b_o = N/2$	13.081 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	9.114 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	508.822 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 509^2 \cdot 1080$$

$$= 22394.541 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 9.11 \cdot \pi \cdot 509 \cdot 3 \cdot 1080$$

$$= 9627.082 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 425^2 \cdot 1080$$

$$= 15616.503 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 22395 - 15617$$

$$= 6778.038 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(22395 + 9627 + 0, 0)$$

$$= 32021.621 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 9.11 \cdot \pi \cdot 509 + 0 \cdot 0 \cdot 0$$

$$= 102430.352 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(W_o + F_a + 4Me/G) / S_b, W_g / S_a$$



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$$= \max((32022 + 0 + 4 \cdot 0 / 509) / 172, 102430 / 172)$$

$$= 58.403 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 28.6 + 6 \cdot 41 / (3 + 0.5)$$

$$= 127.436 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 578 \cdot \sin(\pi / 16)$$

$$= 112.733 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 113 / (2 \cdot 28.6 + 41) }, 1)$$

$$= 1.0717$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	58.403	75.148	
Radial Distance between Hub and Bolts:	38.100	46.074	
Radial Distance between Bolts and Edge:	28.575	28.575	
Circ. Spacing between the Bolts:	63.500	112.733	127.436

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 75.1 \cdot 172 / (68.9 \cdot \pi (527 + 475))$$

$$= 5.956 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 75.1$$

$$= 131798.58 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 32022 - 22395$$

$$= 9627.08 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (578 - 509) / 2$$

$$= 34.5140 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((578 - 425) / 2 + 34.5)$$

$$= 55.4945 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (578 - 425 - 30.4) / 2$$

$$= 61.2745 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	15617.	61.2745	1.0717	1026.



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Face Pressure, Mt	6778.	55.4945	1.0717	403.
Gasket Load, Mg	9627.	34.5140	1.0717	356.
Gasket Seating, Matm	131799.	34.5140	1.0717	4875.

Total Moment for Operation, Mo 1785. Kg-m.
 Total Moment for Gasket seating, Mg 4875. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 82.581 mm
 Hub Ratio, h/h0 = HL / H0 1.212
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.894

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.734
 Factor V per 4.16.5 0.155
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 635/425
 = 1.494

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1)) / (K - 1)
 = (0.66845 + (5.7169(1.49² • log10(1.49))) / (1.49² - 1)) / (1.49 - 1)
 = 5.005

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.49²(1 + 8.55246 • log10(1.49)) - 1) / ((1.0472 + 1.9448 • 1.49²) • (1.49 - 1))
 = 1.713

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
 = (1.49²(1 + 8.55246 • log10(1.49)) - 1) / (1.36136(1.49² - 1) • (1.49 - 1))
 = 5.500

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
 = (1.49² + 1) / (1.49² - 1)
 = 2.621

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 5.5 • 16² • 82.6 / 0.16
 = 752659.938 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.73/82.6
 = 0.009 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (41 • 0.0089 + 1) / 1.71 + 41³/752660
 = 0.888

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1•1785/425)/(0.89•30.4²)
 = 50.18 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)



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$$= (1 \cdot 4875 / 425) / (0.89 \cdot 30.4^2)$$

$$= 137.08 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.49 \cdot 1785 / 425) / (0.89 \cdot 41^2)$$

$$= 41.00 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.49 \cdot 4875 / 425) / (0.89 \cdot 41^2)$$

$$= 112.00 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (5 \cdot 1785 / (41^2 \cdot 425)) - 2.62 \cdot 41$$

$$= 15.15 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (5 \cdot 4875 / (41^2 \cdot 425)) - 2.62 \cdot 112$$

$$= 41.38 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (50.2 + \max(41, 15.2)) / 2$$

$$= 45.59 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (137 + \max(112, 41.4)) / 2$$

$$= 124.54 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 32022 / 75.1$$

$$= 41.79 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (102430 / 75.1)$$

$$= 133.67 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	50.18	198.75	137.08	207.00
Radial Flange	41.00	132.50	112.00	138.00
Tangential Flange	15.15	132.50	41.38	138.00
Maximum Average	45.59	132.50	124.54	138.00
Bolting	41.79	172.00	133.67	172.00

Minimum Required Flange Thickness 41.300 mm

Estimated M.A.W.P. (Operating) 3139 KPa.
 Estimated Finished Weight of Flange at given Thk. 89.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 199.7 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.17, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C



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Flg Calc [Int P]: FLANGE Flng: 18 9:42am Sep 9,2026

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	41.91	207.00	112.40	207.00
Radial Flange	36.15	138.00	96.95	138.00
Tangential Flange	12.96	138.00	34.74	138.00
Maximum Average	39.03	138.00	104.67	138.00
Bolting	41.79	172.00	133.67	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 37.948 mm

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 Internal Pressure Calculations: Step: 6 9:42am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Dish End		1010.3	14	3	1050	138	1
LHS Ch Shell		1010.3	14	3	1050	138	2
LHS Ch Flg		1010.5	115	3	1050	138	3
LHS Shell Flg		1090.5	115	3	1050	132.5	4
Main Shell		1090.3	14	3	1050	137	5
RHS Dish End		1090.3	14	3	1050	137	6
70	80	1080	52	3	418.9	132.5	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Dish End		1000	1436	2231	8.5	6.83964	1
LHS Ch Shell		1000	2829	3622	14	6.88286	2
LHS Ch Flg		1000	2520	3041	107	103.556	3
LHS Shell Flg		1080	2363	3041	107	103.556	4
Main Shell		1080	2808	3622	14	7.22252	5
RHS Dish End		1080	1426	2231	8.5	7.17416	6
70	80	1080	3139	3818	44	41.3004	7

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1426 KPa.
 Shell Side MAPnc = 1610 KPa.
 Channel Side MAWP = 1070 KPa.
 Channel Side MAPnc = 1610 KPa.

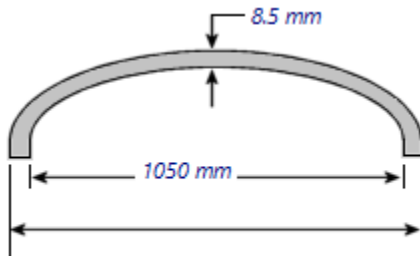
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:			
Design Pressure P	1010.321 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	50.800 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1050.000 mm	Actual Stress	96.4 N./mm ²
Spec. Min. Thk t,e, or sa	8.5000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	14317.2 cm ²	SG of Contents	1.000
Element Volume	195555.6 cm ³	Weight of Contents	195.4 Kg
Empty Weight	165.4 Kg	Operating Weight	360.9 Kg
		Comp. All. Ope. Stress	111.3 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1010 \cdot 1056 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1010)$$

$$= 3.8396 + 3.0000 = 6.8396 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.322 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 5.5) / (0.99 \cdot 1056 + 0.2 \cdot 5.5)$$

$$= 1447 - 10.3 = 1436 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8.5) / (1 \cdot 1050 + 0.2 \cdot 8.5)$$

$$= 2231 \text{ KPa.}$$



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 Element: LHS Dish End Step: 7 9:42am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1010 \cdot (0.99 \cdot 1056 + 0.2 \cdot 5.5)) / (2 \cdot 1 \cdot 5.5)$$

$$= 96.369 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (1010 \cdot 528) / (138 \cdot 1 - 0.6 \cdot 1010) + 3$$

$$= 6.883 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 10.322 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11)$$

$$= 2839 - 10.3 = 2829 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1056 / (2 \cdot 266)))^2 / 6$$

$$= 0.992488$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.660 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8.5, tr = 4.07, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.74$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14, tr = 4.11, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.37$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1069.581 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	8.500 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.739	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

EXTERNAL PRESSURE RESULTS for: LHS Dish End

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.



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Element: LHS Dish End Step: 7 9:42am Sep 9,2026

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
5.500	1067.00	194.00	0.0007159	71.58

$$\text{MAEP} = B / (K0 * \text{Do} / t) = 71.6 / (0.9 * 194) = 410 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.763	1067.00	386.22	0.0003596	35.95

$$\text{MAEP} = B / (K0 * \text{Do} / t) = 36 / (0.9 * 386) = 103 \text{ KPa.}$$

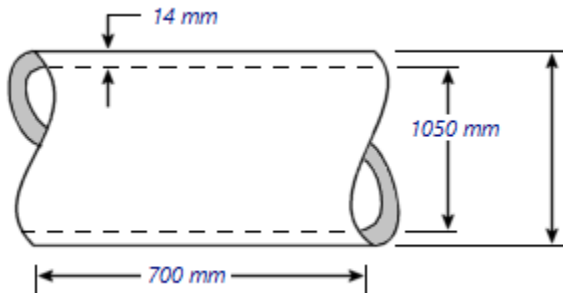
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 FileName : 173-C-_CS_Channel side-Final Page 49 of 200
 Element: LHS Ch Shell Step: 8 9:42am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1010.321 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1050.000 mm	Actual Stress	49.1 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	23706.5 cm ²	SG of Contents	1.000
Element Volume	606239.7 cm ³	Weight of Contents	605.9 Kg
Empty Weight	253.9 Kg	Operating Weight	859.7 Kg
		Comp. All. Ope. Stress	130.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1010 \cdot 528) / (138 \cdot 1 - 0.6 \cdot 1010) \\
 &= 3.8829 + 3.0000 = 6.8829 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.322 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11) \\
 &= 2839 - 10.3 = 2829 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 14) / (525 + 0.6 \cdot 14) \\
 &= 3622 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1010 \cdot (528 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 49.104 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 173-C-_CS_Channel side-Final Page 50 of 200
 Element: LHS Ch Shell Step: 8 9:42am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.316 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 4.11$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.37$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1069.581 KPa.
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.374	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1078.00	978.30	98.00	0.9075	0.0015393	97.46

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 97.5) / (3 \cdot 98) = 1326 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.313	1078.00	978.30	325.35	0.9075	0.0002524	25.24

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 25.2) / (3 \cdot 325) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1078.00	23402.17	98.00	21.7089	0.0001164	11.64

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 11.6) / (3 \cdot 98) = 158 \text{ KPa.}$$

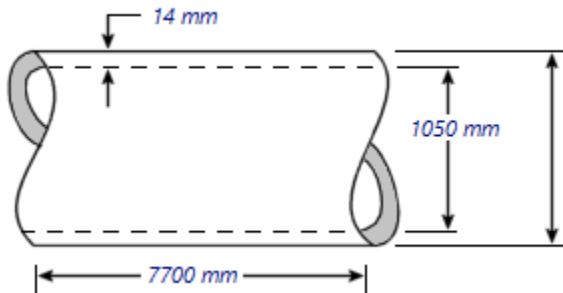
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 Element: Main Shell Step: 9 9:42am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1090.322 KPa.	Int. Design Temperature	275 °C
Material	SA-516 70	Ext. Design Temperature	275 °C
Length	7700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1050.000 mm	Actual Stress	53.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	260771.1 cm ²	SG of Contents	1.000
Element Volume	3988329.0 cm ³	Weight of Contents	3985.9 Kg
Empty Weight	2792.7 Kg	Operating Weight	6778.6 Kg
		Comp. All. Ope. Stress	96.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1090 \cdot 528) / (137 \cdot 1 - 0.6 \cdot 1090)$$

$$= 4.2225 + 3.0000 = 7.2225 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.322 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11)$$

$$= 2819 - 10.3 = 2808 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14) / (525 + 0.6 \cdot 14)$$

$$= 3622 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1090 \cdot (528 + 0.6 \cdot 11)) / (1 \cdot 11)$$

$$= 52.993 \text{ N./mm}^2$$



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Element: Main Shell Step: 9 9:42am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.316 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 5.49$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

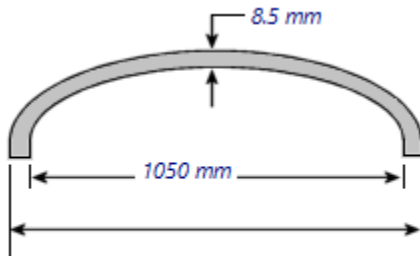
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1425.956 KPa.
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-48 °C
Stress/Thk. Ratio	0.499	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:							
Design Pressure	P	1090.322 KPa.		Int. Design Temperature		275 °C	
Material		SA-516 70		Ext. Design Temperature		275 °C	
Length		50.800 mm		External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	3.0000 mm		Allowable Stress (ope)	S	137.0 N./mm ²	
Ext. Corr. All.	ce	0.0000 mm		Allowable Stress (amb)	Sa	138.0 N./mm ²	
Inside Diameter	Di	1050.000 mm		Actual Stress		104.0 N./mm ²	
Spec. Min. Thk t,e, or sa		8.5000 mm		Spec. Nominal Thk.		14.000 mm	
Surface Area		14317.2 cm ²		SG of Contents		1.000	
Element Volume		195555.6 cm ³		Weight of Contents		195.4 Kg	
Empty Weight		165.4 Kg		Operating Weight		360.9 Kg	
				Comp. All. Ope. Stress		96.7 N./mm ²	
				Aspect Ratio		2.00	
Radiography Information:							
User Defined				User Defined			
Circ. Joint Efficiency		100 %		Long. Joint Eff.		E or v 100 %	

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1090 \cdot 1056 \cdot 0.99) / (2 \cdot 137 \cdot 1 - 0.2 \cdot 1090)$$

$$= 4.1742 + 3.0000 = 7.1742 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 10.322 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137 \cdot 1 \cdot 5.5) / (0.99 \cdot 1056 + 0.2 \cdot 5.5)$$

$$= 1436 - 10.3 = 1426 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8.5) / (1 \cdot 1050 + 0.2 \cdot 8.5)$$

$$= 2231 \text{ KPa.}$$



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 Element: RHS Dish End Step: 10 9:42am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1090 \cdot (0.99 \cdot 1056 + 0.2 \cdot 5.5)) / (2 \cdot 1 \cdot 5.5)$$

$$= 104.000 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (1090 \cdot 528) / (137 \cdot 1 - 0.6 \cdot 1090) + 3$$

$$= 7.223 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 10.322 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (137 \cdot 1 \cdot 11) / (528 + 0.6 \cdot 11)$$

$$= 2819 - 10.3 = 2808 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1056 / (2 \cdot 266)))^2 / 6$$

$$= 0.992488$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.660 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8.5, tr = 5.42, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14, tr = 5.49, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.5$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1425.956 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	8.500 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.986	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-30 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1853.742 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1404.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2082.111 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1568.551 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1544.400 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1544.400 KPa.
App 27-4	= M.A.W.P.	1425.956 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1390.455 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2082.111 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1176.539 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1069.581 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	199.6	234.0	0.853	2092.40
LHS Ch Shell	101.7	234.0	0.435	2092.40
Main Shell	101.7	234.0	0.435	2092.40
RHS Dish End	199.6	234.0	0.853	2092.40

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	138.00	1.000
N3-Shell Inlet	Nozzle	118.00	118.00	1.000
N3-Shell Inlet	Pad	138.00	137.00	1.007
N4-Shell Inlet	Nozzle	118.00	118.00	1.000
N4-Shell Inlet	Pad	138.00	137.00	1.007
N5-Shell Outlet	Nozzle	118.00	118.00	1.000
N5-Shell Outlet	Pad	138.00	137.00	1.007

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Shell Flg	138.00	132.50	1.041
Main Shell	138.00	137.00	1.007



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RHS Dish End	138.00	137.00	1.007
	138.00	132.50	1.041

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	74.15	1.551
Tubesheet Material	138.00	132.50	1.041

Minimum 1.041

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	30.51	216.00	0.141
N2-Tube Outlet	30.51	216.00	0.141
N3-Shell Inlet	29.22	216.00	0.135
N4-Shell Inlet	29.22	216.00	0.135
N5-Shell Outlet	28.97	216.00	0.134

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1853.742 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1404.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2082.111 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1568.551 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1544.400 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1544.400 KPa.
App 27-4	= M.A.W.P.	1425.956 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1390.455 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2082.111 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1176.539 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1069.581 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1080 • 1
 = 1404.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1000 • 1
 = 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	125.0	234.0	0.534	1310.29
LHS Ch Shell	63.7	234.0	0.272	1310.29
Main Shell	68.7	234.0	0.294	1414.29
RHS Dish End	134.9	234.0	0.577	1414.29

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	138.00	1.000
N3-Shell Inlet	Nozzle	118.00	118.00	1.000
N3-Shell Inlet	Pad	138.00	137.00	1.007
N4-Shell Inlet	Nozzle	118.00	118.00	1.000
N4-Shell Inlet	Pad	138.00	137.00	1.007
N5-Shell Outlet	Nozzle	118.00	118.00	1.000
N5-Shell Outlet	Pad	138.00	137.00	1.007

Minimum

1.000



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Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Shell Flg	138.00	132.50	1.041
Main Shell	138.00	137.00	1.007
RHS Dish End	138.00	137.00	1.007
	138.00	132.50	1.041

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	74.15	1.551
Tubesheet Material	138.00	132.50	1.041

Minimum 1.041

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	19.10	216.00	0.088
N2-Tube Outlet	19.10	216.00	0.088
N3-Shell Inlet	19.75	216.00	0.091
N4-Shell Inlet	19.75	216.00	0.091
N5-Shell Outlet	19.58	216.00	0.091

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		1404.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1050.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1050.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	668
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	7350.00 mm
Straight Tube Length (bet. inner tubsht faces) L		7263.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	35.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	505.621 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	7204.8 cm ²
Perimeter of Tube Layout	Cp	3196.3 mm



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Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	87.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1230.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	1226.00 mm
Step 1 Depth on the Tubeside	ht1	0.00 mm
Step 2 Diameter on the Tubeside	dt2	1226.00 mm
Step 2 Depth on the Tubeside	ht2	0.00 mm
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1226.00 mm
Step 1 Depth on the Shellside	hs1	0.00 mm
Step 2 Diameter on the Shellside	ds2	1226.00 mm
Step 2 Depth on the Shellside	hs2	0.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		1230.00 mm

Note: Tubesheet diameter is now: 1230.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	1099.745 mm
Area of the Untubed Lanes	AL	526.9 cm ²

Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	103.0000 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	1230.000 mm
Flange Inside Diameter	B	1050.000 mm
Flange Face Outside Diameter	Fod	1125.000 mm
Flange Face Inside Diameter	Fid	1050.000 mm
Gasket Outside Diameter	Go	1114.000 mm
Gasket Inside Diameter	Gi	1082.000 mm
Small end Hub thk.	g0	14.0000 mm
Large end Hub thk.	g1	20.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	1102.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:



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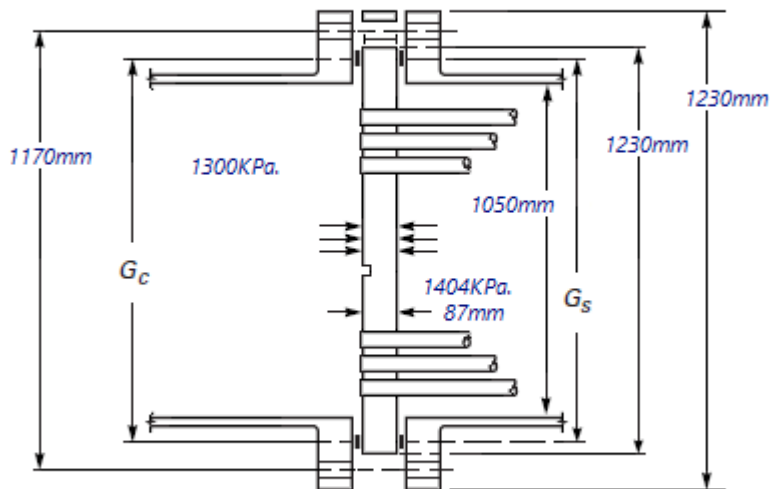
Diameter of Bolt Circle	C	1170.000 mm
Nominal Bolt Diameter	dB	28.5750 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	44
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	161880.42 Kg
Alternate Flange Seating Bolt Load, Wm2	215021.02 Kg
Alternate Flange Design Bolt Load, W	362446.06 Kg

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 55.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
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Elastic Mod. at Ambient Temperature 21.1 °C 0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1300 \cdot (25.4/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 1300)$
 $= 0.1429 + 0.0000 = 0.1429 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 21.11 °C
 Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7768.62	15.39	50.0000	0.0046419	81.86

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 81.9) / (3 \cdot 15.4) = 7090 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5606	25.40	7768.62	45.31	50.0000	0.0005359	51.21

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 51.2) / (3 \cdot 45.3) = 1507 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5606 mm
Allowable Internal Pressure at Corroded thickness	15758.77 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	7090.01 KPa.
Required External Design Pressure	1507.00 KPa.
Required Thickness due to Shell Side pressure	0.5606 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	16.000 mm
Basic Gasket Width,	$b0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \sqrt{b0} \cdot 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	1099.750 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 506 + 25.4$
 $= 1036.642 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (35 - 25.4) / 35$



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= 0.274

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

= G_s / D_o (Configurations d, e, f)
 = 1100/1037
 = 1.0609

Ratio [pc]:

= G_c / D_o (Configurations d)
 = 1100/1037
 = 1.0609

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

= l_{tx} / h
 = 0/87
 = 0.000 (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d*]:

= $\max(d_t - 2t_t (E_t/E) \cdot (S_t/S) \cdot (p) ; d_t - 2t_t)$
 = $\max(25.4 - 2 \cdot 1.65 (195117184/202701232) \cdot (115/138) \cdot (0) ; 25.4 - 2 \cdot 1.65)$
 = 25.4000 mm

Effective tube pitch [p*]:

= $p / \sqrt{ (1 - 4 \cdot \min(A_l ; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$
 = $35 / \sqrt{ (1 - 4 \cdot \min(527 ; 4 \cdot 1037 \cdot 35) / (\pi \cdot 1037^2)) }$
 = 36.1465 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$
 = $(36.1 - 25.4) / 36.1$
 = 0.2973

$h/p = 2.485714$; $\mu^* = 0.297304$; $E^* = 56273968$ KPa.
 $E^*/E = 0.277620$; $v^* = 0.355697$

Note: As h/p (2.486) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

= A / D_o
 = 1230/1037
 = 1.1865

Coefficient [F]:

= $(1 - v^*) / E^* \cdot (E \cdot \ln(K))$
 = $(1 - 0.36) / 56273968 \cdot (202701232 \cdot \ln(1.19))$
 = 0.3969

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

= $D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$
 = $1037^2/16 \cdot [(1.06 - 1) \cdot (1.06^2 + 1) \cdot 1404 - (1.06 - 1) \cdot (1.06^2 + 1) \cdot 1300]$
 = 904808.3125 KPa·mm²

Moment acting on unperforated tubesheet rim [M*]



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$$\begin{aligned}
 &= Mts + W * (Gc - Gs) / (2 * \pi * Do) \\
 &= 904808 + 362446 * (1100 - 1100) / (2 * \pi * 1037) \\
 &= 904808.3125 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Note: W is the maximum of the bolt loads between the shell and channel sides.*

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned}
 &= ((M*) - Do^2/32 * F(Ps - Pt)) / (1 + F) \\
 &= ((904808) - 1037^2/32 * 0.4 (1404 - 1300)) / (1 + 0.4) \\
 &= -344655.0938 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 * (3 + \nu *) * (Ps - Pt) \\
 &= -344655 + 1037^2/64 * (3 + 0.36) * (1404 - 1300) \\
 &= 5515301.5000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(| Mp | ; | Mo |) \\
 &= \max(| -344655 | ; | 5515301 |) \\
 &= 5515301.5000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - hg')^2) \\
 &= 6 * 5515301 / (0.3 (87 - 8)^2) \\
 &= 17.8357 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 * \beta T * Sy \\
 &= 2 * 0.95 * 250 \\
 &= 474.99 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - hg')^2) \\
 &= 6 * 5416983 / (0.3 (23.2 - 8)^2) \\
 &= 473.7275 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 23.2 + 0 + 0 \\
 &= 23.1916 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(23.2 ; 0.45) \\
 &= 23.1916 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= | Ps - Pt | \\
 &= | 1404 - 1300 | \\
 &= 104.000 \text{ KPa.}
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 * \mu * h / Do * 0.8 * \beta T * Sy \\
 &= 4 * 0.27 * 87 / 1037 * 0.8 * 0.95 * 250 \\
 &= 17.494 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.



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Minimum required thickness for Shear [HreqS]:
 $= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$
 $= 1 / (4 \cdot 0.27 \cdot 190) \cdot (4 \cdot 7205 / 3196) \cdot | 1404 - 1300 | + 0$
 $= 0.4499 \text{ mm}$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:
 $= \pi (do - t) t$
 $= \pi (25.4 - 1.65) 1.65$
 $= 1.231 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (do - t) S$
 $= \pi \cdot 1.65 (25.4 - 1.65) 161$
 $= 2027.372 \text{ Kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$
 $= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$
 $= 1923.2015 \text{ Kgf}$

Weld Strength Factor [fw]:
 $= S / Sw \text{ where } Sw = \min (S \text{ and } St)$
 $= 161 / 161$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$
 $= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$
 $= 1244.4244 \text{ Kgf}$

Minimum Length of Weld [ar]:
 $= 2 (\sqrt{ ((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd) - 0.75 \cdot do })$
 $= 2 (\sqrt{ ((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1) - 0.75 \cdot 25.4 })$
 $= 2.141 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
 Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 2027 is $\geq$ tube strength 2027 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	17.8 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	1.0 $\leq$	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	23.192	87.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok



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U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	62.130 ...	223	475	12 190	Ps+Pt D1	Ok
D2uc	66.276 15.068	258	475	14 190	Ps+Pt D2	Ok
D3uc	23.192 ...	18	475	1 190	Ps+Pt D3	Ok
Max:	66.2764 15.068 mm	0.544	0.075	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	66.276	87.000	Ok
Tubesheet thickness flanged extension :	15.068	55.000	Ok
Tube thickness :	0.561	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculations: Step: 13 9:42am Sep 9,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	1067	5.5	0.00071592	71.5761
20	30	978.3	1078	11	0.0015393	97.4564
30	40	No Calc	...	104	No Calc	No Calc
40	50	No Calc	...	104	No Calc	No Calc
50	60	7978.3	1078	11	0.0001604	14.7438
60	70	No Calc	1067	5.5	0.00071592	61.912
70	80	No Calc	...	41	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	8.5	5.76267	103.419	410
20	30	14	6.31334	103.419	1326
30	40	107	103.556	103.419	No Calc
40	50	107	103.556	...	No Calc
50	60	14	No Calc	...	201
60	70	8.5	No Calc	...	355
70	80	44	41.3004	...	No Calc

Minimum

201

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	978.3	23402.2	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	7978.3	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	165.447	195556	129.994	198673	16.5447
20	30	253.878	606240	200.038	613188	25.3878
30	40	301.868	129318	290.484	129645	30.1868
40	50	301.868	129318	290.484	129645	30.1868
50	60	2792.66	3988329	2200.42	4064759	279.266
60	70	165.447	195556	129.994	198673	16.5447
70	80	100.536	21169.6	95.8493	21567.6	10.0536
Total		4081	5265485.50	3337	5356152.00	408



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	195.436	-87.5	0.36335E-04	...	Liquid: 10
20	Liqd	605.87	350	...	...	Liquid: 20
20	Nozl	36.6461	350	634.538	...	N1-Tube Inlet
20	Nozl	36.6461	350	-634.538	...	N2-Tube Outlet
20	Wght	300	350	...	...	Partition weigh
30	Liqd	129.239	70	-9.99997	...	Liquid: 30
40	Liqd	129.239	70	-9.99997	...	Liquid: 40
40	Insl	8.0127	52	...	...	Insulation
50	Sadl	234.616	1540	750.5	...	Left Saddle Flo
50	Sadl	234.616	6160	750.5	...	RH Fixed saddle
50	Liqd	3985.9	3850	...	...	Liquid: 50
50	Insl	1004.24	3850	...	...	Insulation
50	Nozl	112.211	1900	686.925	...	N3-Shell Inlet
50	Nozl	112.211	5700	686.925	...	N4-Shell Inlet
50	Nozl	336.877	3800	-753.6	...	N5-Shell Outlet
50	Wght	250	1540	-1000	...	Fireproofing we
50	Wght	250	6160	1000	...	Fireproofing we
60	Liqd	195.436	138.3	0.36335E-04	...	Liquid: 60
60	Insl	97.5756	25.25	...	...	Insulation
30	FTsh	630.278	183.85	...	...	U-Tube
30	Tube	5688.88	3815.35	...	...	Tubes
40	Baf1	45.9167	672	...	...	Baffles
40	Baf1	45.9167	1344	...	...	Baffles
40	Baf1	45.9167	2016	...	...	Baffles
40	Baf1	45.9167	2688	...	...	Baffles
40	Baf1	45.9167	3360	...	...	Baffles
40	Baf1	45.9167	4032	...	...	Baffles
40	Baf1	45.9167	4704	...	...	Baffles
40	Baf1	45.9167	5376	...	...	Baffles
40	Baf1	45.9167	6048	...	...	Baffles
40	Baf1	46.3413	6720	...	...	Baffles

Total Weight of Each Detail Type:

Saddles 469.2
 Liquid 5241.1



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 Element and Detail Weights: Step: 14 9:42am Sep 9,2026

Insulation	1109.8
Nozzles	634.6
Weights	800.0
Exchanger Components	6778.7
Liquid in Tubes	2086.0
Sum of the Detail Weights	17119.5 kg.

Note:
 Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	4379.3	4379.3	4379.3	4379.3	4379.3	4379.3
Saddles	469.2	469.2	469.2	469.2	469.2	469.2
Nozzles	634.6	634.6	634.6	634.6	634.6	634.6
Wld Weights	300.0	300.0	300.0	300.0	300.0	300.0
Exchanger	6778.7	6778.7	6778.7	6778.7	6778.7	6778.7
Insulation	...	...	...	...	1109.8	1109.8 *
Empty Weights	...	...	...	500.0	500.0	...
Test Weights	...	500.0	...	...	...	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	5241.1
Tube Ope Lqd	...	...	...	...	...	2086.0
Test Liquid	...	5262.3	...	...	...	...
Tube Tst Lqd	...	2086.0	...	...	...	...
Totals	12561.9	20410.1	12561.9	13061.9	14171.7	21498.8

Field Installation Options:
 * Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	12561.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	20410.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	12561.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	13061.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	14171.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	21498.8 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	20685.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	20902.3 kg.

Exchanger Tube Data		
Volume of Exchanger tubes :	2087269.9 cm ³	
Weight of Ope Liq in tubes :	2086.0 kg.	
Weight of Water in tubes :	2086.0 kg.	

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
------	----	---	--



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Element and Detail Weights:

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10	20	14317.2	13626.7
20	30	23706.5	23090.7
30	40	8797.09	4882.04
40	50	8797.09	4882.04
50	60	260771	253998
60	70	14317.2	13626.7
70	80	4353.12	2001.34

Total	335059.3	316107.2 cm ²
Total	360.7	340.3 ft ²

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	2935.4	1069.6	1820.4	-204.7
N2-Tube Outlet	1820.4	1960.0	82	150	GR 1.1	2935.4	1069.6	1820.4	-204.7
N3-Shell Inlet	4095.0	5100.0	275	300	GR 1.1	2169.2	3973.4	4095.0	3973.4
N4-Shell Inlet	4095.0	5100.0	275	300	GR 1.1	2214.1	3928.5	4095.0	3928.5
N5-Shell Outle	4095.0	5100.0	275	300	GR 1.1	1795.5	4095.0	4095.0	4095.0

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 3928.483 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1069.581 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	248.539	1.20	807	1530	1069.6	1069.6	Pass
N2-Tube Outlet	248.539	1.20	807	1530	1069.6	1069.6	Pass
N3-Shell Inlet	359.785	0.50	1816	2295	1426.0	3973.4	Pass
N4-Shell Inlet	357.251	0.50	1816	2295	1426.0	3928.5	Pass
N5-Shell Outlet	500.888	0.50	4087	3442	1426.0	4095.0	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(807) + 4 \cdot \text{abs}(1530) 248.539 \leq \pi \cdot 248.539^3 [(1820.434 - 1069.581) + 1.200 \cdot 1820.434]$$

$$= 0.14438E+05 \leq 0.14438E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 14437.8 / (\pi \cdot 248.539^3))$$

$$= \min(1820.434, 1069.581)$$

$$= 1069.581 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(807) + 4 \cdot \text{abs}(1530) 248.539 \leq \pi \cdot 248.539^3 [(1820.434 - 1069.581) + 1.200 \cdot 1820.434]$$

$$= 0.14438E+05 \leq 0.14438E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 14437.8 / (\pi \cdot 248.539^3))$$

$$= \min(1820.434, 1069.581)$$

$$= 1069.581 \text{ KPa.}$$



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 Nozzle Flange MAWP: Step: 15 9:42am Sep 9,2026

Evaluating Flange on Nozzle [N3-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) 359.785 \leq \pi \cdot 359.785^3 [(4095.047 - 1425.956) + .500 \cdot 4095.047]$$

$$= 0.32366E+05 \leq 0.70374E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4095.047, 0.500 \cdot 4095.047 + 4095.047 - 32365.5 / (\pi \cdot 359.785^3))$$

$$= \min(4095.047, 3973.377)$$

$$= 3973.377 \text{ KPa.}$$

Evaluating Flange on Nozzle [N4-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) 357.251 \leq \pi \cdot 357.251^3 [(4095.047 - 1425.956) + .500 \cdot 4095.047]$$

$$= 0.32342E+05 \leq 0.68898E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4095.047, 0.500 \cdot 4095.047 + 4095.047 - 32342.3 / (\pi \cdot 357.251^3))$$

$$= \min(4095.047, 3928.483)$$

$$= 3928.483 \text{ KPa.}$$

Evaluating Flange on Nozzle [N5-Shell Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(4087) + 4 \cdot \text{abs}(3442) 500.888 \leq \pi \cdot 500.888^3 [(4095.047 - 1425.956) + .500 \cdot 4095.047]$$

$$= 0.72289E+05 \leq 0.18989E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N5-Shell Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4095.047, 0.500 \cdot 4095.047 + 4095.047 - 72288.7 / (\pi \cdot 500.888^3))$$

$$= \min(4095.047, 4347.038)$$

$$= 4095.047 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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 Wind Load Calculation: Step: 16 9:42am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	8050 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.95) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.518
Structure Height to Diameter ratio	2.106

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000 + 8050 = 9050 \text{ mm}$
 $= 29.7 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (29.7 ft.) $> 15 \text{ ft.}$
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(29.7/900)^{2/9.5}$
 $= 0.98$

Type of Hill: No Hill



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Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = (1 + K1 • K2 • K3)²
 = (1 + 0 • 0 • 0)²
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
 = max(16, 0.00256 • 0.98 • 1 • 1 • 1 • 60.4²)
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 = qz • G • Cf • WindArea
 = 16 • 0.85 • 0.52 • 3.14
 = 22.2 lbs. [10.1] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Dish End	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
LHS Ch Shell	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
LHS Ch Flg	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
LHS Shell Flg	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
Main Shell	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
RHS Dish End	9050.0	0.000	0.000	0.000	0.980	1.000	78.120
Node 70 to 80	8122.0	0.000	0.000	0.000	0.958	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition wei	0.00	9050.00	78.12	0.00
Fireproofing	0.00	8050.00	78.12	0.00
Fireproofing	0.00	10050.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	9050	1280.4	2921.48	78.12	11.0631
20	30	9050	1293.6	9055.2	78.12	34.2904
30	40	9050	1260	1764	78.12	6.67995
40	50	9050	1349.14	1888.8	78.12	7.15254
50	60	9050	1533.6	118087	78.12	447.175
60	70	9050	1518.98	4463.46	78.12	16.9023
70	80	8122.04	502.68	724.241	78.12	2.74257

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = 0.020(50.75)
 = 0.067 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.067$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 21030 \cdot 1.25$
 = 1236.538 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 21030$
 = 659.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	525	2628.69	154.567



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Earthquake Load Calculation:

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20	30	525	2628.69	154.567
30	40	525	2628.69	154.567
40	50	525	2628.69	154.567
50	Sad1	525	2628.69	154.567
Sad1	60	525	2628.69	154.567
50	60	525	2628.69	154.567
60	70	525	2628.69	154.567

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 9:42am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4974.150 mm
Center of Gravity of the Liquid	4206.527 mm
Center of Gravity of the Insulation	5286.428 mm
Center of Gravity of the Nozzles	4401.726 mm
Center of Gravity of the Added Weights (Operating)	3259.144 mm
Center of Gravity of the Added Weights (Empty)	3259.144 mm
Center of Gravity of the Tubesheet(s)	934.650 mm
Center of Gravity of the Tubes	4566.150 mm
Center of Gravity of the Baffles	4682.943 mm
Center of Gravity of Bare Shell New and Cold	4044.274 mm
Center of Gravity of Bare Shell Corroded	3949.853 mm
Vessel CG in the Operating Condition	4251.509 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4145.815 mm
Vessel CG in the Test Condition	4204.682 mm

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	532.00	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	7890.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1540.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	266.75	mm
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		6800.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		3989.88	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.13	-234.00	
Long. Stress at Top of Midspan	1.13	137.00	
Long. Stress at Bottom of Midspan	1.13	137.00	
Long. Stress at Top of Saddles	10.06	137.00	
Long. Stress at Bottom of Saddles	-5.58	-234.00	
Long. Stress at Bottom of Saddles	5.58	137.00	
Tangential Shear in Shell	3.59	109.60	
Circ. Stress at Horn of Saddle	6.19	205.50	
Circ. Compressive Stress in Shell	0.24	137.00	

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 6800 • 1.5 • 1000/4620
 = 2207.8 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 1782 + \max(2208, 0, 0, 0) \\
 &= 3989.9 \text{ Kg}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4224.49	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	10200.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(1050/2 + 14 + 10)\sin(120/2), 2 \cdot 532) \\
 &= 950.896 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -3990 \cdot 1540 [1 - (1 - 1540/7891 + (532^2 - 267^2)/(2 \cdot 1540 \cdot 7891))/(1 + (4 \cdot 267)/(3 \cdot 7891))] \\
 &= -1361.2 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L] \\
 &= 3990 \cdot 7891/4 [(1 + 2(532^2 - 267^2)/(7891^2))/(1 + (4 \cdot 267)/(3 \cdot 7891)) - 4 \cdot 1540/7891] \\
 &= 1438.2 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 0 \cdot 532/(2 \cdot 14) - 1438/(\pi \cdot 532^2 \cdot 14) \\
 &= -1.13 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 0 \cdot 532/(2 \cdot 14) + 1438/(\pi \cdot 532^2 \cdot 14) \\
 &= 1.13 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 0 \cdot 532/(2 \cdot 14) - 1361/(0.11 \cdot \pi \cdot 532^2 \cdot 14) \\
 &= 10.06 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 0 \cdot 532/(2 \cdot 14) + 1361/(0.19 \cdot \pi \cdot 532^2 \cdot 14) \\
 &= -5.58 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 3990(7891 - 2 \cdot 1540)/(7891 + (4 \cdot 267/3)) \\
 &= 2327.6 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$



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$$= 1.17 \cdot 2328 / (532 \cdot 14)$$

$$= 3.59 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(532 \cdot 14)}$$

$$= 67.315 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(532 \cdot 14)}, 2 \cdot 1540)$$

$$= 509.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 3990 \cdot 0.1 / (510(14 + 1 \cdot 10))$$

$$= -0.24 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -3990 / (4(14 + 1 \cdot 10)510) - 3 \cdot 0.053 \cdot 3990 / (2(14 + 1 \cdot 10)^2)$$

$$= -6.19 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	475.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(525 \cdot 14)} + 510)14^3) / 12$$

$$= 14.720 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.7 + 90.1(186 - 7)^2$$

$$= 28862.814 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	643.7	14.0	7.0	90.1	63086.7	0.289E+05
Wearplate	510.0	10.0	19.0	51.0	96900.0	0.142E+05
Web	10.0	431.0	239.5	43.1	1032245.0	0.791E+04
BasePlate	400.0	20.0	465.0	80.0	3719999.8	0.623E+05
Totals	...	...	...	264.2	4912231.5	0.113E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4912231 / 264$$

$$= 185.912 \text{ mm}$$



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Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3990$$

$$= 812.0265 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.4574 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 525 \cdot \sin(120/2) / 1.05$$

$$= 565.828 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 459.4768 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.7392 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3990 + 235) \cdot 400) / (4 \cdot 1000 \cdot 173)}$$

$$= 8.468 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (525 + 14 + 10)$$

$$= 950.896 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 951 / (3 - 1)$$

$$= 475.448 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 475 \cdot 400 / 2$$

$$= 950.896 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4224 / (1000 \cdot 400)$$

$$= 1.056 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 951 \cdot 1.06$$

$$= 1004.263 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 23.8$$

$$= 60.272 \text{ cm}^2$$



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Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1004/60.3$$

$$= 1.634 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 10$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4395/60.3)}$$

$$= 85.388 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (525 + 14 + 10) - 20$$

$$= 705.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 706/85.4$$

$$= 8.262$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 10200 / (2 \cdot 1000) \cdot 475 \cdot 706/2$$

$$= 855.360 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.26 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.26)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.26) / (8 \cdot 123) - (8.26^3) / (8 \cdot 123^3))$$

$$= 153.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.63/153 + (855 \cdot 188/43945308) / 173$$

$$= 0.217$$



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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	475.4	10.0	0.0	47.5	0.0	3.96
Totals	...	...	...	84.0	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/84$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(549/2)(20 + 0 + 0) - 0$$

$$= 431.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4398/84)}$$

$$= 72.342 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 431/72.3$$

$$= 5.958$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 10200 / (2 \cdot 1000)$$

$$= 5.100 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 5.1 \cdot 475 \cdot 431$$

$$= 1045.103 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1045/234582$$

$$= 43.690 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.96 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.96)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (5.96) / (8 \cdot 123) - (5.96^3) / (8 \cdot 123^3))$$

$$= 154.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 2.34/154 + (1045 \cdot 188/4398) / 173$$

$$= 0.267$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7



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Bolt Allowable Stress	Stba	130.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	4224.5 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	10200.0 Kgf
Maximum Transverse Force	Ft	0.0 Kgf
Maximum Longitudinal Force	F1	10200.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $10200 / (4.87 \cdot 4)$
 = 51.4 N./mm². Must be less than 78.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $0 / (4.87 \cdot 4)$
 = 0.0 N./mm². Must be less than 78.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = $(QL - Q0) / (Stba \cdot Rnbolts)$
 = $(10200 - 4224) / (130 \cdot 4)$
 = 1.1270 cm²

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $10200 / (78 \cdot 4)$
 = 3.2061 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1000 \cdot 0 + 0$
 = 0.00 Kg-m.

Eccentricity (e):
 = $Rmom / Q0$
 = $0 / 4224$
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[1.13, 3.21, 0]$
 = 3.2061 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	533.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	7890.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1540.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	268.25	mm
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		6800.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		3580.43	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-1.29	-234.00
Long. Stress at Top of Midspan		1.29	137.00
Long. Stress at Bottom of Midspan		1.29	137.00
Long. Stress at Top of Saddles		11.43	137.00
Long. Stress at Bottom of Saddles		-6.34	-234.00
Long. Stress at Bottom of Saddles		6.34	137.00
Tangential Shear in Shell		4.09	109.60
Circ. Stress at Horn of Saddle		7.16	205.50
Circ. Compressive Stress in Shell		0.26	137.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 6800 • 1.5 • 1000/4620
 = 2207.8 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 1373 + \max(2208, 0, 0, 0) \\
 &= 3580.4 \text{ Kg}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3815.05	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	10200.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(1050/2 + 14 + 10)\sin(120/2), 2 \cdot 534) \\
 &= 950.896 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -3580 \cdot 1540 [1 - (1 - 1540/7891 + (534^2 - 268^2)/(2 \cdot 1540 \cdot 7891))/(1 + (4 \cdot 268)/(3 \cdot 7891))] \\
 &= -1222.4 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L] \\
 &= 3580 \cdot 7891/4 [(1 + 2(534^2 - 268^2)/(7891^2))/(1 + (4 \cdot 268)/(3 \cdot 7891)) - 4 \cdot 1540/7891] \\
 &= 1289.2 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 0 \cdot 534/(2 \cdot 11) - 1289/(\pi \cdot 534^2 \cdot 11) \\
 &= -1.29 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 0 \cdot 534/(2 \cdot 11) + 1289/(\pi \cdot 534^2 \cdot 11) \\
 &= 1.29 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 0 \cdot 534/(2 \cdot 11) - 1222/(0.11 \cdot \pi \cdot 534^2 \cdot 11) \\
 &= 11.43 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 0 \cdot 534/(2 \cdot 11) + 1222/(0.19 \cdot \pi \cdot 534^2 \cdot 11) \\
 &= -6.34 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 3580(7891 - 2 \cdot 1540)/(7891 + (4 \cdot 268/3)) \\
 &= 2088.2 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$



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$$= 1.17 \cdot 2088 / (534 \cdot 11)$$

$$= 4.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(534 \cdot 11)}$$

$$= 59.753 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(534 \cdot 11)}, 2 \cdot 1540)$$

$$= 494.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 3580 \cdot 0.1 / (495(11 + 1 \cdot 10))$$

$$= -0.26 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -3580 / (4(11 + 1 \cdot 10)495) -$$

$$3 \cdot 0.053 \cdot 3580 / (2(11 + 1 \cdot 10)^2)$$

$$= -7.16 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	475.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(528 \cdot 11)} + 510)11^3) / 12$$

$$= 6.975 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 6.98 + 69.2(199 - 5.5)^2$$

$$= 25839.752 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	628.9	11.0	5.5	69.2	38047.7	0.258E+05
Wearplate	510.0	10.0	16.0	51.0	81600.0	0.170E+05
Web	10.0	431.0	236.5	43.1	1019314.9	0.729E+04
BasePlate	400.0	20.0	462.0	80.0	3695999.5	0.555E+05
Totals	...	...	...	243.3	4834962.0	0.106E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4834962 / 243$$

$$= 198.743 \text{ mm}$$



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Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3580$$

$$= 728.6962 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.4105 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 528 \cdot \sin(120/2) / 1.05$$

$$= 563.347 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 410.5173 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.7574 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(3580 + 235)400 / (4 \cdot 1000 \cdot 173)}$$

$$= 8.047 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (525 + 14 + 10)$$

$$= 950.896 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 951 / (3 - 1)$$

$$= 475.448 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 475 \cdot 400 / 2$$

$$= 950.896 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3815 / (1000 \cdot 400)$$

$$= 0.954 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 951 \cdot 0.95$$

$$= 906.929 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 23.8$$

$$= 60.272 \text{ cm}^2$$



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Compressive Stress [Sc]:

$$= P/A_r$$

$$= 907/60.3$$

$$= 1.476 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 10$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4395/60.3)}$$

$$= 85.388 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1000 - \cos(120/2) (525 + 14 + 10) - 20$$

$$= 705.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 706/85.4$$

$$= 8.262$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 10200 / (2 \cdot 1000) \cdot 475 \cdot 706/2$$

$$= 855.360 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.26 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.26)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.26) / (8 \cdot 123) - (8.26^3) / (8 \cdot 123^3))$$

$$= 153.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.48/153 + (855 \cdot 188/43945308) / 173$$

$$= 0.216$$



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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	475.4	10.0	0.0	47.5	0.0	3.96
Totals	...	...	...	84.0	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/84$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(549/2)(20 + 0 + 0) - 0$$

$$= 431.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4398/84)}$$

$$= 72.342 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 431/72.3$$

$$= 5.958$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 10200 / (2 \cdot 1000)$$

$$= 5.100 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 5.1 \cdot 475 \cdot 431$$

$$= 1045.103 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1045/234582$$

$$= 43.690 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.96 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.96)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (5.96) / (8 \cdot 123) - (5.96^3) / (8 \cdot 123^3))$$

$$= 154.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 2.12/154 + (1045 \cdot 188/4398) / 173$$

$$= 0.266$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
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Bolt Allowable Stress	Stba	130.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	3815.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	10200.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	10200.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $10200 / (4.87 \cdot 4)$
 = 51.4 N./mm². Must be less than 78.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $0 / (4.87 \cdot 4)$
 = 0.0 N./mm². Must be less than 78.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = $(QL - Q0) / (Stba \cdot Rnbolts)$
 = $(10200 - 3815) / (130 \cdot 4)$
 = 1.2042 cm²

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $10200 / (78 \cdot 4)$
 = 3.2061 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1000 \cdot 0 + 0$
 = 0.00 Kg-m.

Eccentricity (e):
 = $Rmom / Q0$
 = $0 / 3815$
 = 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 = $\max[1.2, 3.21, 0]$
 = 3.2061 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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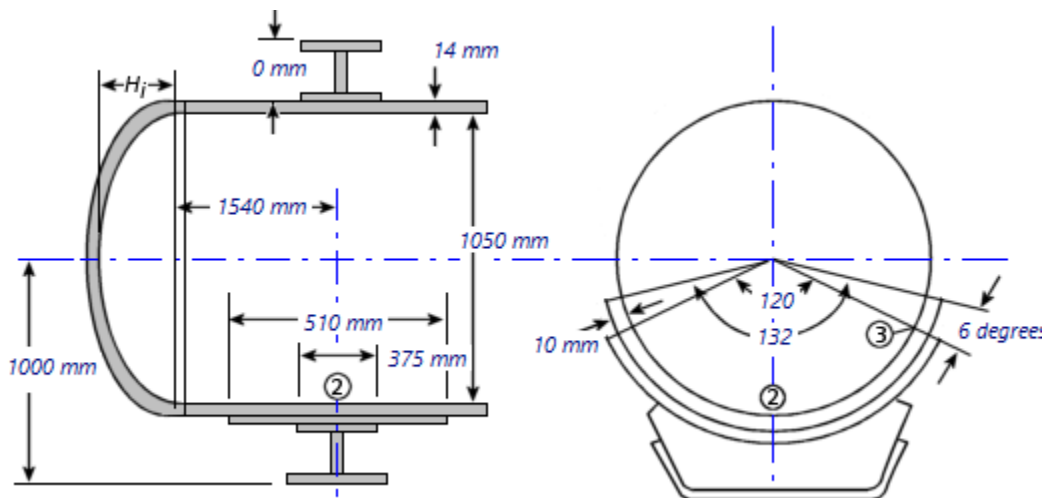
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	533.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	7890.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1540.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		132.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		16192.16	Kgf
Pressure used in Analysis	P	1085.146	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	19.04	137.00
Long. Stress at Bottom of Midspan	33.59	137.00
Long. Stress at Top of Saddles	69.10	137.00
Long. Stress at Bottom of Saddles	2.60	137.00
Tangential Shear in Shell	19.31	109.60
Circ. Stress at Horn of Saddle	32.37	171.25
Circ. Compressive Stress in Shell	1.16	137.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (526/2 + 0) \cdot 1000/951$$

$$= 829.8 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(113, 0) \cdot 1000/4620$$

$$= 24.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$$

$$= \max(1237, 0) \cdot 1000/4620$$

$$= 267.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3 (1237/2 + 0) \cdot 1000/951$$

$$= 1950.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 14242 + \max(24.4, 830, 268, 1951)$$

$$= 16192.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(1.08 \cdot 1.2)^2)$$

$$= 1107.542 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	16426.78	Kgf
Transverse Shear Load	618.27	Kgf
Longitudinal Shear Load	1236.54	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1050/2 + 14 + 10) \sin(120/2), 2 \cdot 534)$$

$$= 950.896 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$$

$$= -16192 \cdot 1540 [1 - (1 - 1540/7891 + (534^2 - 0^2)/$$

$$(2 \cdot 1540 \cdot 7891)) / (1 + (4 \cdot 0)/(3 \cdot 7891))]$$

$$= -4574.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot \text{L} / 4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/3\text{L}) - 4a/L)$$

$$= 16192 \cdot 7891 / 4 ((1 + 2(534^2 - 0^2)/(7891^2)) / (1 + (4 \cdot 0)/$$



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$$(3 \cdot 7891) - 4 \cdot 1540 / 7891 \\ = 7298.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 534 / (2 \cdot 11) - 7299 / (\pi \cdot 534^2 \cdot 11) \\ = 19.04 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 534 / (2 \cdot 11) + 7299 / (\pi \cdot 534^2 \cdot 11) \\ = 33.59 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 534 / (2 \cdot 11) - 4575 / (0.11 \cdot \pi \cdot 534^2 \cdot 11) \\ = 69.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1085 \cdot 534 / (2 \cdot 11) + 4575 / (0.19 \cdot \pi \cdot 534^2 \cdot 11) \\ = 2.60 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 16192 (7891 - 2 \cdot 1540) / (7891 + (4 \cdot 0 / 3)) \\ = 9871.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 9872 / (534 \cdot 11) \\ = 19.31 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{534 \cdot 11} \\ = 59.753 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(375 + 1.56 \cdot \sqrt{534 \cdot 11}, 2 \cdot 1540) \\ = 494.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(137 / 137, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 16192 \cdot 0.1 / (495(11 + 1 \cdot 10)) \\ = -1.16 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -16192 / (4(11 + 1 \cdot 10)495) - 3 \cdot 0.053 \cdot 16192 / (2(11 + 1 \cdot 10)^2) \\ = -32.37 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 4620.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.13139\text{E-}04 \cdot 4620 (275 - 21.1)$$



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= 15.411 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	210.0	N./mm ²
Height of Web at Center	Hw,c	475.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(528 \cdot 11)} + 510)11^3)/12$
 $= 6.975 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 6.98 + 69.2(199 - 5.5)^2$
 $= 25839.752 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	628.9	11.0	5.5	69.2	38047.7	0.258E+05
Wearplate	510.0	10.0	16.0	51.0	81600.0	0.170E+05
Web	10.0	431.0	236.5	43.1	1019314.9	0.729E+04
BasePlate	400.0	20.0	462.0	80.0	3695999.5	0.555E+05
Totals	...	...	...	243.3	4834962.0	0.106E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 4834962/243$
 $= 198.743 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 16192$
 $= 3295.4563 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 1.8563 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 126.0000 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1000 - 528 \cdot \sin(120/2) / 1.05$
 $= 563.347 \text{ mm}$

Bending Moment, M = Fh * d = 1856.5237 Kg-m.



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Bending Stress, $S_b = (M \cdot C_1 / I) = 3.4254 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 140.0000 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(16192 + 235) \cdot 400) / (4 \cdot 1000 \cdot 140)}$
 $= 18.580 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (525 + 14 + 10)$
 $= 950.896 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 951 / (3 - 1)$
 $= 475.448 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 475 \cdot 400 / 2$
 $= 950.896 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 16427 / (1000 \cdot 400)$
 $= 4.107 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 951 \cdot 4.11$
 $= 3905.040 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 23.8$
 $= 60.272 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 3905 / 60.3$
 $= 6.354 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375 - 10$
 $= 365.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 60.3$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4395/60.3}$
 = 85.388 mm

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1000 - \cos(120/2) (525 + 14 + 10) - 20$
 = 705.500 mm

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 199943008/210)}$
 = 137.095

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 706/85.4$
 = 8.262

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 1237 / (2 \cdot 1000) \cdot 475 \cdot 706/2$
 = 103.695 Kg-m.

Compressive Allowable, $KL/r < Cc$ (8.26 < 137) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (8.26)^2 / (2 \cdot 137^2)) 210 / (5/3 + 3 \cdot (8.26) / (8 \cdot 137) - (8.26)^3 / (8 \cdot 137^3))$
 = 124.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 6.35/124 + (104 \cdot 188/43945308) / 140$
 = 0.082

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	475.4	10.0	0.0	47.5	0.0	3.96
Totals	...	...	...	84.0	...	0.440E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0/84$
 = 0.000 mm

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 375/2$
 = 187.500 mm

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 1000 - \cos(549/2) (20 + 0 + 0) - 0$
 = 431.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(4398/84)}$$

$$= 72.342 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 431/72.3$$

$$= 5.958$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1237 / (2 \cdot 1000)$$

$$= 0.618 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.62 \cdot 475 \cdot 431$$

$$= 126.697 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 127/234582$$

$$= 5.297 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.96 < 137$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.96)^2 / (2 \cdot 137^2)) 210 / (5/3 + 3 \cdot (5.96) / (8 \cdot 137) - (5.96^3) / (8 \cdot 137^3))$$

$$= 124.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 9.12/125 + (127 \cdot 188/4398) / 140$$

$$= 0.111$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	137.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	14476.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	267.6 Kg
Maximum Transverse Force	Ft	618.3 Kg
Maximum Longitudinal Force	F1	1236.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1237 / (4.87 \cdot 4)$$

$$= 6.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 618 / (4.87 \cdot 4)$$

$$= 3.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 = $1237 / (82.8 \cdot 4)$
 = 0.3661 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1000 \cdot 618 + 0$
 = 618.28 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = $618 / 14476$
 = 42.71 mm < $B_{\text{plen}} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 = $\max[0, 0.37, 0]$
 = 0.3661 cm²

MDMT Calculations for: Left Saddle Flo, Saddle to Wearplate Junction, Curve: D

Govrn. thk, $t_g = 10$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = 33 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	533.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	7890.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1540.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	268.25	mm
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		137.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 9508.65 Kg
 Pressure used in Analysis P 1085.146 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	22.90	137.00
Long. Stress at Bottom of Midspan	29.73	137.00
Long. Stress at Top of Saddles	56.68	137.00
Long. Stress at Bottom of Saddles	9.49	137.00
Tangential Shear in Shell	10.85	109.60
Circ. Stress at Horn of Saddle	19.01	171.25
Circ. Compressive Stress in Shell	0.68	137.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (526/2 + 0) \cdot 1000/951$
 $= 829.8 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(113, 0) \cdot 1000/4620$
 $= 24.4 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1237, 0) \cdot 1000/4620$
 $= 267.6 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1237/2 + 0) \cdot 1000/951$
 $= 1950.6 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 7558 + \max(24.4, 830, 268, 1951)$
 $= 9508.7 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (1.08 \cdot 1.2)^2)$
 $= 1107.542 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	9743.27	Kg
Transverse Shear Load	618.27	Kg
Longitudinal Shear Load	1236.54	Kg

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1050/2 + 14 + 10) \sin(120/2), 2 \cdot 534)$
 $= 950.896 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434



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$$K7p = 0.0434$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \frac{(1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))}{(1 + (4hm)/3L)} \right]$$

$$= -9509 \cdot 1540 \left[1 - \frac{(1 - 1540/7891 + (534^2 - 268^2)/(2 \cdot 1540 \cdot 7891))}{(1 + (4 \cdot 268)/(3 \cdot 7891))} \right]$$

$$= -3246.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 9509 \cdot 7891/4 \left(\frac{(1 + 2(534^2 - 268^2)/(7891^2))}{(1 + (4 \cdot 268)/(3 \cdot 7891))} - 4 \cdot 1540/7891 \right)$$

$$= 3423.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 1085 \cdot 534/(2 \cdot 11) - 3424/(\pi \cdot 534^2 \cdot 11)$$

$$= 22.90 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 1085 \cdot 534/(2 \cdot 11) + 3424/(\pi \cdot 534^2 \cdot 11)$$

$$= 29.73 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 1085 \cdot 534/(2 \cdot 11) - 3246/(0.11 \cdot \pi \cdot 534^2 \cdot 11)$$

$$= 56.68 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 1085 \cdot 534/(2 \cdot 11) + 3246/(0.19 \cdot \pi \cdot 534^2 \cdot 11)$$

$$= 9.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 9509 (7891 - 2 \cdot 1540)/(7891 + (4 \cdot 268/3))$$

$$= 5545.8 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 5546/(534 \cdot 11)$$

$$= 10.85 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{534 \cdot 11}$$

$$= 59.753 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{534 \cdot 11}, 2 \cdot 1540)$$

$$= 494.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(137/137, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$



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$$= -0.76 \cdot 9509 \cdot 0.1 / (495 (11 + 1 \cdot 10))$$

$$= -0.68 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -9509 / (4(11 + 1 \cdot 10)495) -$$

$$3 \cdot 0.053 \cdot 9509 / (2(11 + 1 \cdot 10)^2)$$

$$= -19.01 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	210.0	N./mm ²
Height of Web at Center	Hw,c	475.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(528 \cdot 11) + 510}) 11^3) / 12$$

$$= 6.975 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 6.98 + 69.2 (199 - 5.5)^2$$

$$= 25839.752 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	628.9	11.0	5.5	69.2	38047.7	0.258E+05
Wearplate	510.0	10.0	16.0	51.0	81600.0	0.170E+05
Web	10.0	431.0	236.5	43.1	1019314.9	0.729E+04
BasePlate	400.0	20.0	462.0	80.0	3695999.5	0.555E+05
Totals	...	...	...	243.3	4834962.0	0.106E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4834962 / 243$$

$$= 198.743 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 9509$$

$$= 1935.2166 \text{ Kgf}$$

Tension Stress, $St = (Fh / As) = 1.0901 \text{ N./mm}^2$



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Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 126.0000 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1000 - 528 \cdot \sin(120/2) / 1.05$
 $= 563.347 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 1090.2209 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 2.0115 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 140.0000 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(9509 + 235) \cdot 400) / (4 \cdot 1000 \cdot 140)}$
 $= 14.309 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) \cdot (525 + 14 + 10)$
 $= 950.896 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 951 / (3 - 1)$
 $= 475.448 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 475 \cdot 400 / 2$
 $= 950.896 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 9743 / (1000 \cdot 400)$
 $= 2.436 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 951 \cdot 2.44$
 $= 2316.208 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 23.8$
 $= 60.272 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 2316 / 60.3$
 $= 3.769 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375 - 10$



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= 365.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/60.3
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4395/60.3}$
 = 85.388 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1000 - cos(120/2)(525 + 14 + 10) - 20
 = 705.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008/210}$
 = 137.095

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 706/85.4
 = 8.262

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 1237 / (2 • 1000) • 475 • 706/2
 = 103.695 Kg-m.

Compressive Allowable, KL/r < Cc (8.26 < 137) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (8.26)^2 / (2 \cdot 137^2)) 210 / (5/3 + 3 \cdot (8.26) / (8 \cdot 137) - (8.26^3) / (8 \cdot 137^3))$
 = 124.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 3.77/124 + (104 • 188/43945308) / 140
 = 0.061

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	475.4	10.0	0.0	47.5	0.0	3.96
Totals	...	...	...	84.0	...	0.440E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/84
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 375/2



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= 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1000 - cos(549/2)(20 + 0 + 0) - 0
 = 431.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4398/84}$
 = 72.342 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 431/72.3
 = 5.958

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 1237/(2 • 1000)
 = 0.618 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.62 • 475 • 431
 = 126.697 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 127/234582
 = 5.297 N./mm²

Compressive Allowable, KL/r < Cc (5.96 < 137) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(5.96)²/(2 • 137²))210/
 (5/3+3•(5.96)/(8• 137)-(5.96³)/(8•137³))
 = 124.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 5.41/125 + (127 • 188/4398)/140
 = 0.081

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	137.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	7792.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	267.6 Kg
Maximum Transverse Force	Ft	618.3 Kg
Maximum Longitudinal Force	F1	1236.5 Kg
Saddle Bolted to Steel Foundation	No	No

Friction force due to thermal growth, [Ff]:

= 4342.9 Kg

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:



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= max(Ff; F_I) / (Bolt Area • Number of Bolts)
 = 4343 / (4.87 • 4)
 = 21.9 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = F_t / (Bolt Area • Number of Bolts)
 = 618 / (4.87 • 4)
 = 3.1 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = F_I / (BoltShearAllowable • Nbolts)
 = 1237 / (82.8 • 4)
 = 0.3661 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • F_t + Sum of X Moments
 = 1000 • 618 + 0
 = 618.28 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 618 / 7793
 = 79.34 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.37, 0]
 = 0.3661 cm²

MDMT Calculations for: RH Fixed saddle, Saddle to Wearplate Junction, Curve: D

Govrn. thk, t_g = 10, c = 3 mm, E* = 1
 Thickness Ratio = t_r • E* / (t_g - c) = 0.5, Temp. Reduction = 33 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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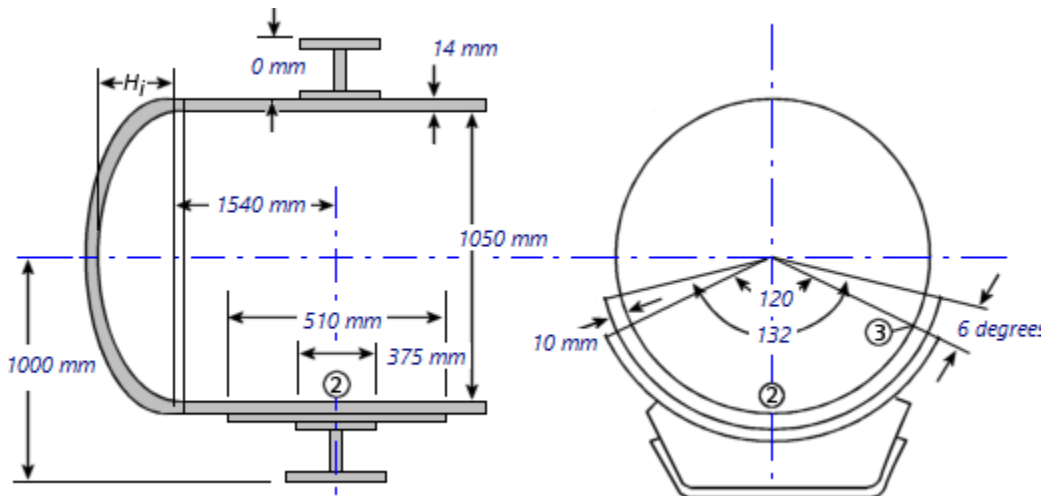
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	533.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	7890.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1540.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	13858.37	Kgf
Pressure used in Analysis	P	1409.229 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	27.95	234.00
Long. Stress at Bottom of Midspan	40.40	234.00
Long. Stress at Top of Saddles	70.79	234.00
Long. Stress at Bottom of Saddles	13.88	234.00
Tangential Shear in Shell	16.53	138.58
Circ. Stress at Horn of Saddle	27.70	350.99
Circ. Compressive Stress in Shell	0.99	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (263/2 + 0) \cdot 1000/951$
 $= 414.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(56.5, 0) \cdot 1000/4620$
 $= 12.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 13443 + \max(12.2, 415, 0, 0)$
 $= 13858.4 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(1.08 \cdot 1.2)^2)$
 $= 1107.542 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	14092.99	Kgf
Transverse Shear Load	131.50	Kgf
Longitudinal Shear Load	56.47	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1050/2 + 14 + 10)\sin(120/2), 2 \cdot 534)$
 $= 950.896 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -13858 \cdot 1540 [1 - (1 - 1540/7891 + (534^2 - 0^2)/(2 \cdot 1540 \cdot 7891))/(1 + (4 \cdot 0)/(3 \cdot 7891))]$
 $= -3915.3 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 $= 13858 \cdot 7891/4 ((1 + 2(534^2 - 0^2)/(7891^2))/(1 + (4 \cdot 0)/(3 \cdot 7891)) - 4 \cdot 1540/7891)$
 $= 6246.6 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 $= 1409 \cdot 534/(2 \cdot 11) - 6247/(\pi \cdot 534^2 \cdot 11)$
 $= 27.95 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$



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$$= 1409 \cdot 534 / (2 \cdot 11) + 6247 / (\pi \cdot 534^2 \cdot 11)$$

$$= 40.40 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1409 \cdot 534 / (2 \cdot 11) - 3915 / (0.11 \cdot \pi \cdot 534^2 \cdot 11)$$

$$= 70.79 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1409 \cdot 534 / (2 \cdot 11) + 3915 / (0.19 \cdot \pi \cdot 534^2 \cdot 11)$$

$$= 13.88 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 13858 (7891 - 2 \cdot 1540) / (7891 + (4 \cdot 0 / 3))$$

$$= 8449.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 8449 / (534 \cdot 11)$$

$$= 16.53 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(534 \cdot 11)}$$

$$= 59.753 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(534 \cdot 11)}, 2 \cdot 1540)$$

$$= 494.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 13858 \cdot 0.1 / (495(11 + 1 \cdot 10))$$

$$= -0.99 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -13858 / (4(11 + 1 \cdot 10)495) -$$

$$3 \cdot 0.053 \cdot 13858 / (2(11 + 1 \cdot 10)^2)$$

$$= -27.70 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	475.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(528 \cdot 11)} + 510)^3) / 12$$

$$= 6.975 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 6.98 + 69.2 (199 - 5.5)^2$$

$$= 25839.752 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	628.9	11.0	5.5	69.2	38047.7	0.258E+05
Wearplate	510.0	10.0	16.0	51.0	81600.0	0.170E+05
Web	10.0	431.0	236.5	43.1	1019314.9	0.729E+04
BasePlate	400.0	20.0	462.0	80.0	3695999.5	0.555E+05
Totals	...	...	...	243.3	4834962.0	0.106E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4834962/243$$

$$= 198.743 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 13858$$

$$= 2820.4795 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.5887 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1000 - 528 \cdot \sin(120/2) / 1.05$$

$$= 563.347 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1588.9414 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.9317 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(13858 + 235) \cdot 400) / (4 \cdot 1000 \cdot 173)}$$

$$= 15.466 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (525 + 14 + 10)$$

$$= 950.896 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 951 / (3 - 1)$$

$$= 475.448 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 475 \cdot 400 / 2$$

$$= 950.896 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 14093 / (1000 \cdot 400)$$

$$= 3.523 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 951 \cdot 3.52$$

$$= 3350.242 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 36.5 + 23.8$$

$$= 60.272 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 3350 / 60.3$$

$$= 5.451 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 375 - 10$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 60.3$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (4395 / 60.3) }$$

$$= 85.388 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1000 - \cos(120 / 2) (525 + 14 + 10) - 20$$

$$= 705.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 706/85.4$$

$$= 8.262$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 56.5 / (2 \cdot 1000) \cdot 475 \cdot 706/2$$

$$= 4.735 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($8.26 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.26)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (8.26) / (8 \cdot 123) - (8.26^3) / (8 \cdot 123^3))$$

$$= 153.3 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.45/153 + (4.74 \cdot 188/43945308) / 173$$

$$= 0.037$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	475.4	10.0	0.0	47.5	0.0	3.96
Totals	...	...	...	84.0	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/84$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1000 - \cos(549/2) (20 + 0 + 0) - 0$$

$$= 431.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4398/84) }$$

$$= 72.342 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 431/72.3$$

$$= 5.958$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 56.5 / (2 \cdot 1000)$$

$$= 0.028 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.028 \cdot 475 \cdot 431$$

$$= 5.786 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 5.79/234582$$

$$= 0.242 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.96 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.96)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (5.96) / (8 \cdot 123) - (5.96^3) / (8 \cdot 123^3))$$

$$= 154.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 7.82/154 + (5.79 \cdot 188/4398) / 173$$

$$= 0.052$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load QO (Weight)	QO	13678.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	12.2 Kg
Maximum Transverse Force	Ft	131.5 Kg
Maximum Longitudinal Force	F1	56.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 56.5 / (4.87 \cdot 4)$$

$$= 0.3 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 132 / (4.87 \cdot 4)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 56.5 / (103 \cdot 4)$$

$$= 0.0134 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1000 \cdot 132 + 0$$

$$= 131.50 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 132/13678$$

$$= 9.61 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.013, 0]
 = 0.0134 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	533.50	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	7890.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1540.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	510.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	268.25	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1000.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		7022.86	Kgf
Pressure used in Analysis	P	1409.229	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	31.65	234.00
Long. Stress at Bottom of Midspan	36.70	234.00
Long. Stress at Top of Saddles	56.60	234.00
Long. Stress at Bottom of Saddles	21.75	234.00
Tangential Shear in Shell	8.01	138.58
Circ. Stress at Horn of Saddle	14.04	350.99
Circ. Compressive Stress in Shell	0.50	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (263/2 + 0) • 1000/951
 = 414.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(56.5, 0) • 1000/4620
 = 12.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



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= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 6608 + max(12.2, 415, 0, 0)
 = 7022.9 Kgf

Longitudinal Wind Force [FI]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (1.08 • 1.2)²)
 = 1107.542 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7257.48	Kgf
Transverse Shear Load	131.50	Kgf
Longitudinal Shear Load	56.47	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1050/2 + 14 + 10)sin(120/2), 2•534)
 = 950.896 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -7023•1540[1 - (1 - 1540/7891 + (534²-268²)/(2•1540•7891))/(1+(4•268)/(3•7891))]
 = -2397.7 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 7023•7891/4((1+2(534²-268²)/(7891²))/(1+(4•268)/(3•7891)) - 4•1540/7891)
 = 2528.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 1409 • 534/(2•11) - 2529/(π •534²•11)
 = 31.65 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 1409 • 534/(2 • 11) + 2529/(π • 534² • 11)
 = 36.70 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 1409•534/(2•11) - 2398/(0.11• π •534²•11)
 = 56.60 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 1409•534/(2•11) + 2398/(0.19• π •534²•11)
 = 21.75 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = Q(L-2a)/(L+(4•hm/3))



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$$= 7023 (7891 - 2 \cdot 1540) / (7891 + (4 \cdot 268 / 3))$$

$$= 4096.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 4096 / (534 \cdot 11)$$

$$= 8.01 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (534 \cdot 11) }$$

$$= 59.753 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{ (534 \cdot 11) }, 2 \cdot 1540)$$

$$= 494.51 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot t_r))$$

$$= -0.76 \cdot 7023 \cdot 0.1 / (495 (11 + 1 \cdot 10))$$

$$= -0.50 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4 (t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2 (t + \eta \cdot t_r)^2)$$

$$= -7023 / (4 (11 + 1 \cdot 10) 495) -$$

$$3 \cdot 0.053 \cdot 7023 / (2 (11 + 1 \cdot 10)^2)$$

$$= -14.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1000.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	475.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (528 \cdot 11) } + 510) 11^3) / 12$$

$$= 6.975 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 6.98 + 69.2 (199 - 5.5)^2$$

$$= 25839.752 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	628.9	11.0	5.5	69.2	38047.7	0.258E+05
Wearplate	510.0	10.0	16.0	51.0	81600.0	0.170E+05
Web	10.0	431.0	236.5	43.1	1019314.9	0.729E+04
BasePlate	400.0	20.0	462.0	80.0	3695999.5	0.555E+05
Totals	...	...	...	243.3	4834962.0	0.106E+06



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 4834962/243$$

$$= 198.743 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 7023$$

$$= 1429.3043 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.8051 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1000 - 528 \cdot \sin(120/2) / 1.05$$

$$= 563.347 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 805.2109 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.4857 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(7023 + 235)400 / (4 \cdot 1000 \cdot 173)}$$

$$= 11.099 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (525 + 14 + 10)$$

$$= 950.896 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 951 / (3 - 1)$$

$$= 475.448 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 475 \cdot 400/2$$

$$= 950.896 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 7257 / (1000 \cdot 400)$$

$$= 1.814 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 951 \cdot 1.81$$

$$= 1725.276 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 36.5 + 23.8
 = 60.272 cm²

Compressive Stress [Sc]:

= P/Ar
 = 1725/60.3
 = 2.807 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 375 - 10
 = 365.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/60.3
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4395/60.3}$
 = 85.388 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1000 - cos(120/2)(525 + 14 + 10) - 20
 = 705.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 706/85.4
 = 8.262

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 56.5 / (2 • 1000) • 475 • 706/2
 = 4.735 Kg-m.

Compressive Allowable, KL/r < Cc (8.26 < 123) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (8.26)² / (2 • 123²)) 260 /
 (5/3 + 3 • (8.26) / (8 • 123) - (8.26³) / (8 • 123³))
 = 153.3 N./mm²



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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.81/153 + (4.74 \cdot 188/43945308) / 173$$

$$= 0.019$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	475.4	10.0	0.0	47.5	0.0	3.96
Totals	...	...	...	84.0	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/84$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1000 - \cos(549/2) (20 + 0 + 0) - 0$$

$$= 431.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4398/84) }$$

$$= 72.342 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 431/72.3$$

$$= 5.958$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 56.5 / (2 \cdot 1000)$$

$$= 0.028 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.028 \cdot 475 \cdot 431$$

$$= 5.786 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5.79/234582$$

$$= 0.242 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($5.96 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (5.96)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (5.96) / (8 \cdot 123) - (5.96^3) / (8 \cdot 123^3))$$

$$= 154.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 4.03/154 + (5.79 \cdot 188/4398) / 173$$

$$= 0.028$$



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Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	130.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	6842.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	12.2 Kg
Maximum Transverse Force	Ft	131.5 Kg
Maximum Longitudinal Force	Fl	56.5 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 56.5 / (4.87 \cdot 4)$
 $= 0.3 \text{ N./mm}^2$. Must be less than 78.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 132 / (4.87 \cdot 4)$
 $= 0.7 \text{ N./mm}^2$. Must be less than 78.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 56.5 / (78 \cdot 4)$
 $= 0.0177 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1000 \cdot 132 + 0$
 $= 131.50 \text{ Kg-m}$.

Eccentricity (e):
 $= Rmom / Q0$
 $= 132 / 6843$
 $= 19.22 \text{ mm} < Bplen/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.018, 0]$
 $= 0.0177 \text{ cm}^2$

Review notes about nozzle loadings and supports in the Warnings report.

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 Nozzle Calcs.: N1-Tube Inlet Nozl: 11 9:43am Sep 9,2026

Input, Nozzle Desc: N1-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1050.00	mm
Design Length of Section	L	978.3000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.80	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

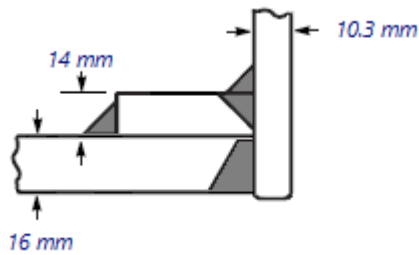
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	170.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	320.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		50.4625	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 8.625 in.
 Actual Thickness Used in Calculation 0.406 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 528 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 3.8430 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1000 \cdot 110 / (118 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.9252 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6299 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	408.9004	mm
Parallel to Vessel Wall, opening length	d	204.4502	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.2810	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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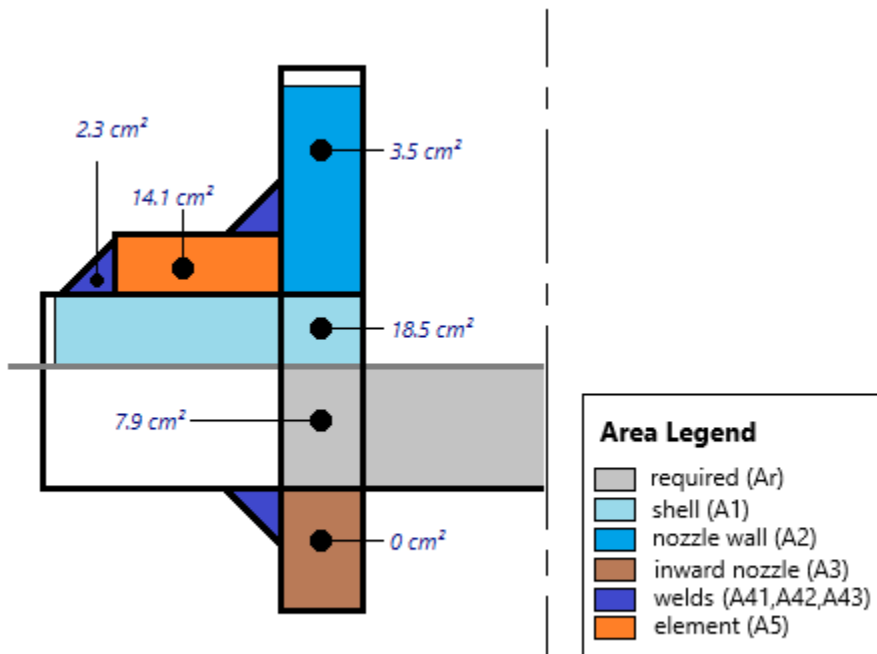


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.939	3.422	NA
Area in Shell	A1	18.527	19.599	NA
Area in Nozzle Wall	A2	3.526	3.689	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	14.129	14.129	NA
TOTAL AREA AVAILABLE	Atot	38.478	39.712	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (204 \cdot 3.84 \cdot 1 + 2 \cdot 7.31 \cdot 3.84 \cdot 1 \cdot (1 - 0.86))$
 $= 7.939 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 204(1 \cdot 13 - 1 \cdot 3.84) - 2 \cdot 7.31$
 $(1 \cdot 13 - 1 \cdot 3.84) \cdot (1 - 0.86)$
 $= 18.527 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.3) \cdot (7.31 - 0.93) \cdot 0.86$



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$$= 3.526 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$
 $= 2.295 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$
 $= (320 - 219) \cdot 14 \cdot 1$
 $= 14.129 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.9252 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 6.8430 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 6.8430 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 3.3959 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 10.1600 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[10.2, \max(6.84, 4.5)]$
 $= 6.8430 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.93, 6.84)$
 $= 6.8430 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 10.312 = 9.023 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 35.4,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 259.6 N./mm ²	Passed
Occasional	: 6.7,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 22.7,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.15$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.15$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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 Nozzle Calcs.: N1-Tube Inlet Nozl: 11 9:43am Sep 9,2026

Govrn. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 14$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.00 / 1960.01 = 0.510$

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 7.3124 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.1187 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$6.5000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (7.94 - 18.5 + 2 \cdot 7.31 \cdot 0.86 \cdot (1 \cdot 13 - 3.84)) \cdot 138)$
 $= \max(0, -13289.08)$ Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.53 + 14.1 + 2.3 - 0 \cdot 0.86) \cdot 138$
 $= 28073.66$ Kgf

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.53 + 0 + 0.86 + (1.63)) \cdot 138$
 $= 8452.53$ Kgf



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Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (3.53 + 0 + 2.3 + 14.1 + (1.63)) \cdot 138$
 $= 30361.26 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$
 $= 20289. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 320 \cdot 12 \cdot 0.49 \cdot 138$
 $= 41590. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 106) \cdot (10.3 - 3) \cdot 0.7 \cdot 118$
 $= 20487. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 219 \cdot 14 \cdot 0.74 \cdot 138$
 $= 50167. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 219 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 46584. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (41590 + 20487) = 62077 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (20289 + 50167 + 46584 + 0) = 117040 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (S_{pew} + T_{ngw} + S_{inw})$
 $= (41590 + 46584 + 0) = 88174 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 62076 Kgf, must exceed W = 0 Kgf	or W1 = 28073 Kgf
Path 2-2 = 117039 Kgf, must exceed W = 0 Kgf	or W2 = 8452 Kgf
Path 3-3 = 88174 Kgf, must exceed W = 0 Kgf	or W3 = 30361 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1562 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 11.5542 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 197.5542 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	219.075	mm
Nozzle Thickness	10.3124	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	320.000	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{541.5(16.0 - 3.0)}$
 $= 495.951 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1083.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	27.000	mm
Nozzle Thickness used	(t) =	7.312	mm
T / t	=	3.692	
d / t	=	29.959	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.281$$

Nr/P	=	0.073
Mr/P	=	0.074



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MØ/P = 0.088
 NØ/P = 0.243
 MrD/Mc = 0.139
 NrDL/Mc = 0.109
 MØd/Mc = 0.211
 NØDL/Mc = 0.196
 MrD/Ml = 0.107
 NrDL/Ml = 0.090
 MØD/Ml = 0.150
 NØDL/Ml = 0.381

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	-9	32	79	-4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-5	21	45	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	13	34	80	18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4



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Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-25	41	76	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20
Inplane Stress Summary	-3	37	62	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	26	42	78	34

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17
Hoop Stress Summary	7	9	34	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	63	-63	63	-63
Axial Membrane (Mc)	0	0	0	0



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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-22	-22	22	22
Axial Bending (ML)	-176	176	176	-176
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-124	102	273	-206
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14

Shear Stress Summary	18	18	10	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	137	105	274	243
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	14	15	28	29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	63	-63	63	-63
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-242	242	242	-242
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-190	168	339	-272
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	14	14	14	14

Shear Stress Summary	10	10	18	18
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	206	168	340	304

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{534.5(16.0 - 3.0)}$$

$$= 494.155 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1069.000 mm
Nozzle Outside Diameter	(d) =	320.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	57.775 mm
T / t	=	0.225
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.714$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.115
M0/P	=	0.035
N0/P	=	0.068
MrD/Mc	=	0.273
NrDL/Mc	=	0.092
M0d/Mc	=	0.082
N0DL/Mc	=	0.085
MrD/Ml	=	0.090
NrDL/Ml	=	0.069
M0D/Ml	=	0.026
N0DL/Ml	=	0.085

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	40	41	40	41
Outplane Stress Summary	39	36	89	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-56	56	56	-56
Inplane Pressure Stress	20	20	20	20



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Inplane Stress Summary	23	13	149	-84
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	41	37	150	106
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary	-15	49	104	-34
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-169	169	169	-169
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	41	40	41

Inplane Stress Summary	-72	145	286	-174
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	72	145	286	175

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-9	-9	17	17	-2	-2	4	4
Circ. Q (SUS)	-21	21	41	-41	-20	20	38	-38
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-2	-2	4	4	-2	-2	10	10
Long. Q (SUS)	-13	13	31	-31	-34	34	54	-54
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	10.3	11.2	36.0	37.0	17.1	18.1	23.2	24.2
Pm+Pl+Q (Total)	13.2	33.2	77.1	18.3	27.2	42.0	74.2	35.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	37.04	207.00	Passed
Pm+Pl+Q (TOTAL)	77.12	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	41	40	41	40	41	40	41
Circ. Pl (SUS)	-2	-2	14	14	-4	-4	14	14
Circ. Q (SUS)	2	-2	34	-34	-108	108	230	-230
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	12	12	-2	-2	14	14
Long. Q (SUS)	5	-5	117	-117	-33	33	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	40.0	41.0	40.0	41.0	40.0	41.0	40.0	41.0
Pm+Pl (SUS)	38.0	39.0	54.0	55.0	36.1	37.1	54.0	55.0



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Pm+Pl+Q (Total)	40.9	37.6	149.1	106.1	72.1	145.0	284.1	175.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	55.05	207.00	Passed
Pm+Pl+Q (TOTAL)	284.09	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-9	-9	17	17	-2	-2	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-19	-19	25	25	-19	-19	25	25
Long. Q (SUS)	-113	113	239	-239	-179	179	305	-305
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	19.0	19.9	36.0	36.5	25.7	26.7	34.1	34.2
Pm+Pl+Q (Total)	135.3	105.0	272.5	241.0	205.2	168.8	338.9	300.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	36.54	177.00	Passed
Pm+Pl+Q (TOTAL)	338.92	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N2-Tube Outlet

From: 20

Pressure for Reinforcement Calculations	P	1010.292	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1050.00	mm
Design Length of Section	L	978.3000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.80	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

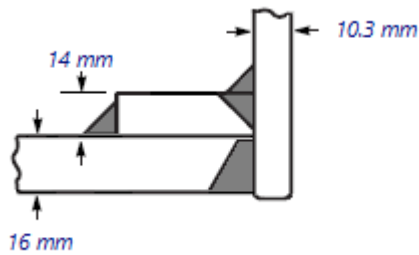
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	170.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	320.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		50.4625	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: N2-Tube Outlet Nozl: 12 9:43am Sep 9,2026



Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 8.625 in.
 Actual Thickness Used in Calculation 0.406 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1010 \cdot 528 / (138 \cdot 1 - 0.6 \cdot 1010)$
 $= 3.8827 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1010 \cdot 110 / (118 \cdot 1 + 0.4 \cdot 1010)$
 $= 0.9347 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6299 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	408.9004	mm
Parallel to Vessel Wall, opening length	d	204.4502	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.2810	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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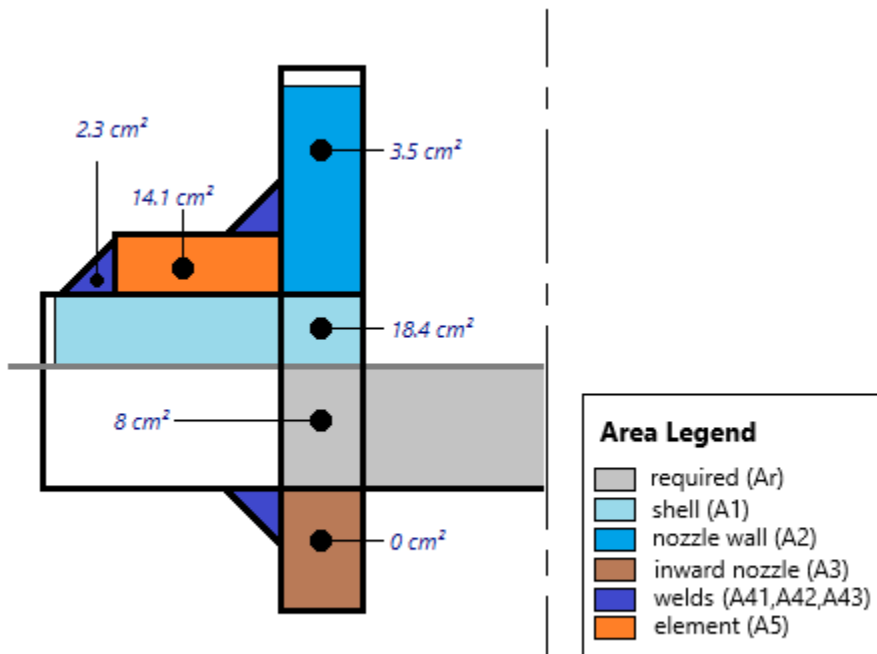


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.021	3.422	NA
Area in Shell	A1	18.447	19.599	NA
Area in Nozzle Wall	A2	3.521	3.689	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	14.129	14.129	NA
TOTAL AREA AVAILABLE	Atot	38.392	39.712	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (204 \cdot 3.88 \cdot 1 + 2 \cdot 7.31 \cdot 3.88 \cdot 1 \cdot (1 - 0.86))$$

$$= 8.021 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 204(1 \cdot 13 - 1 \cdot 3.88) - 2 \cdot 7.31$$

$$(1 \cdot 13 - 1 \cdot 3.88) \cdot (1 - 0.86)$$

$$= 18.447 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 32.3) \cdot (7.31 - 0.93) \cdot 0.86$$



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$$= 3.521 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$
 $= 2.295 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$
 $= (320 - 219) \cdot 14 \cdot 1$
 $= 14.129 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.9347 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 6.8827 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 6.8827 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 3.3959 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 10.1600 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[10.2, \max(6.88, 4.5)]$
 $= 6.8827 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(3.93, 6.88)$
 $= 6.8827 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 10.312 = 9.023 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 35.5,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 259.5 N./mm ²	Passed
Occasional	: 6.8,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 22.7,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.16$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.16$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, $t_g = 14$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.3$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 9.02$, $t_r = 0.93$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1010.29 / 1960.01 = 0.515$

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds T_{min} 7.3124 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.1187 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$6.5000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (8.02 - 18.4 + 2 \cdot 7.31 \cdot 0.86 \cdot (1 \cdot 13 - 3.88)) \cdot 138)$
 $= \max(0, -13067.48)$ Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (3.52 + 14.1 + 2.3 - 0 \cdot 0.86) \cdot 138$
 $= 28066.29$ Kgf

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (3.52 + 0 + 0.86 + (1.63)) \cdot 138$
 $= 8445.16$ Kgf



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Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (3.52 + 0 + 2.3 + 14.1 + (1.63)) \cdot 138$$

$$= 30353.88 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 20289. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 320 \cdot 12 \cdot 0.49 \cdot 138$$

$$= 41590. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 106) \cdot (10.3 - 3) \cdot 0.7 \cdot 118$$

$$= 20487. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 219 \cdot 14 \cdot 0.74 \cdot 138$$

$$= 50167. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 219 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 46584. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (41590 + 20487) = 62077 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (20289 + 50167 + 46584 + 0) = 117040 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (41590 + 46584 + 0) = 88174 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 62076	Kgf, must exceed W = 0 Kgf	or W1 = 28066 Kgf
Path 2-2 = 117039	Kgf, must exceed W = 0 Kgf	or W2 = 8445 Kgf
Path 3-3 = 88174	Kgf, must exceed W = 0 Kgf	or W3 = 30353 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1562 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 11.5542 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 197.5542 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	219.075	mm
Nozzle Thickness	10.3124	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	320.000	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1010.29	KPa.
Radial Load	P	-1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{541.5(16.0 - 3.0)}$
 $= 495.951 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1083.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	27.000	mm
Nozzle Thickness used	(t) =	7.312	mm
T / t	=	3.692	
d / t	=	29.959	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.281$$

Nr/P	=	0.073
Mr/P	=	0.074



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MØ/P = 0.088
 NØ/P = 0.243
 MrD/Mc = 0.139
 NrDL/Mc = 0.109
 MØd/Mc = 0.211
 NØDL/Mc = 0.196
 MrD/Ml = 0.107
 NrDL/Ml = 0.090
 MØD/Ml = 0.150
 NØDL/Ml = 0.381

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	-9	32	80	-3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-5	21	45	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	13	34	80	17

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4



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Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-25	41	76	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	9	-9	9	-9
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-29	29	29	-29
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20
Inplane Stress Summary	-3	38	62	-12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	25	42	78	34

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17
Hoop Stress Summary	8	9	34	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	63	-63	63	-63
Axial Membrane (Mc)	0	0	0	0



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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-22	-22	22	22
Axial Bending (ML)	-176	176	176	-176
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-124	102	273	-206
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14

Shear Stress Summary	18	18	10	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	137	105	274	243
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	14	15	28	29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	63	-63	63	-63
Axial Membrane (Mc)	-22	-22	22	22
Axial Bending (Mc)	-242	242	242	-242
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-190	168	339	-272
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	14	14	14	14

Shear Stress Summary	10	10	18	18
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	206	169	341	304

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{534.5(16.0 - 3.0)}$$

$$= 494.155 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1069.000 mm
Nozzle Outside Diameter	(d) =	320.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	57.775 mm
T / t	=	0.225
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.714$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.061
Mr/P	=	0.115
M0/P	=	0.035
N0/P	=	0.068
MrD/Mc	=	0.273
NrDL/Mc	=	0.092
M0d/Mc	=	0.082
N0DL/Mc	=	0.085
MrD/Ml	=	0.090
NrDL/Ml	=	0.069
M0D/Ml	=	0.026
N0DL/Ml	=	0.085

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	40	41	40	41
Outplane Stress Summary	39	36	90	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-56	56	56	-56
Inplane Pressure Stress	20	20	20	20



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Inplane Stress Summary	23	13	150	-84
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	41	37	150	106
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary	-15	50	104	-34
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-169	169	169	-169
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	41	40	41

Inplane Stress Summary	-72	145	286	-174
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	72	146	286	174

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-9	-9	17	17	-2	-2	4	4
Circ. Q (SUS)	-21	21	41	-41	-20	20	38	-38
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-2	-2	4	4	-2	-2	10	10
Long. Q (SUS)	-13	13	31	-31	-34	34	54	-54
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	10.3	11.2	36.0	37.0	17.1	18.1	23.2	24.2
Pm+Pl+Q (Total)	13.2	33.2	77.1	18.3	27.2	42.0	74.2	35.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	37.04	207.00	Passed
Pm+Pl+Q (TOTAL)	77.12	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	41	40	41	40	41	40	41
Circ. Pl (SUS)	-2	-2	14	14	-4	-4	14	14
Circ. Q (SUS)	2	-2	34	-34	-108	108	230	-230
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	12	12	-2	-2	14	14
Long. Q (SUS)	5	-5	117	-117	-33	33	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	40.0	41.0	40.0	41.0	40.0	41.0	40.0	41.0
Pm+Pl (SUS)	38.0	39.0	54.0	55.0	36.1	37.1	54.0	55.0



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Pm+Pl+Q (Total)	40.9	37.6	149.1	106.1	72.1	145.0	284.1	175.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	55.05	207.00	Passed
Pm+Pl+Q (TOTAL)	284.09	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-9	-9	17	17	-2	-2	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-19	-19	25	25	-19	-19	25	25
Long. Q (SUS)	-113	113	239	-239	-179	179	305	-305
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	19.0	19.9	36.0	36.5	25.7	26.7	34.1	34.2
Pm+Pl+Q (Total)	135.3	105.0	272.5	241.0	205.2	168.8	338.9	300.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	36.54	177.00	Passed
Pm+Pl+Q (TOTAL)	338.92	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N3-Shell Inlet

From: 50

Pressure for Reinforcement Calculations	P	1080.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1050.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3024.15	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

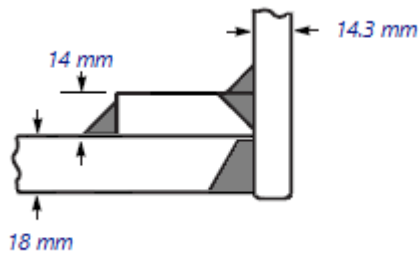
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	473.8500	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1080 \cdot 528 / (137 \cdot 1 - 0.6 \cdot 1080)$
 $= 4.1824 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1080 \cdot 162 / (118 \cdot 1 + 0.4 \cdot 1080)$
 $= 1.4767 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	602.6008 mm
Parallel to Vessel Wall, opening length	d	301.3004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.861$

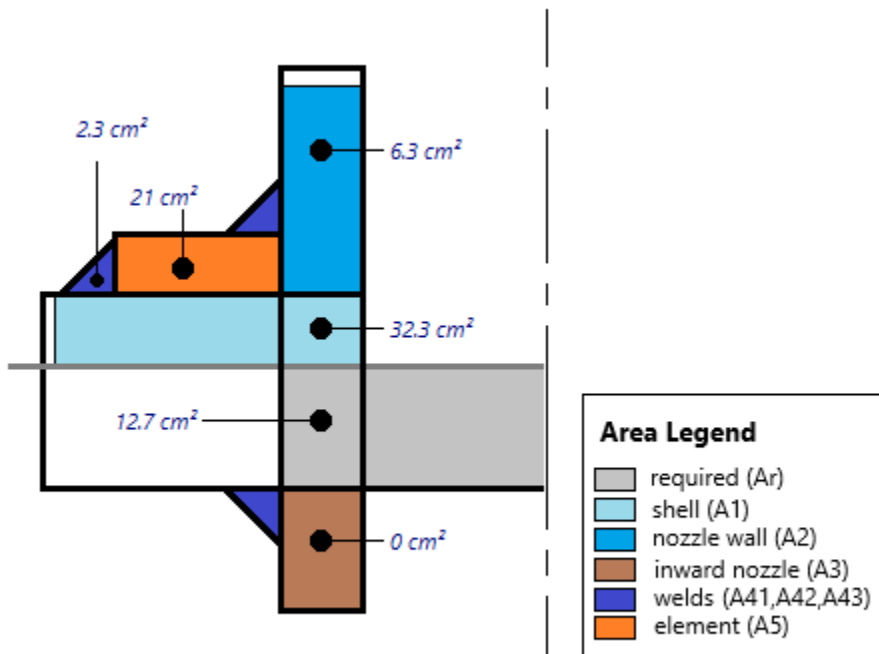


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.732	NA	NA
Area in Shell	A1	32.255	NA	NA
Area in Nozzle Wall	A2	6.329	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.301	NA	NA
Area in Element	A5	21.000	NA	NA
TOTAL AREA AVAILABLE	Atot	61.886	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (301 \cdot 4.18 \cdot 1 + 2 \cdot 11.3 \cdot 4.18 \cdot 1 \cdot (1 - 0.86))$
 $= 12.732 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 301(1 \cdot 15 - 1 \cdot 4.18) - 2 \cdot 11.3$
 $(1 \cdot 15 - 1 \cdot 4.18) \cdot (1 - 0.86)$
 $= 32.255 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (11.3 - 1.48) \cdot 0.86$



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= 6.329 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$
 = 2.301 cm²

Area Available in Element [A5]:
 = $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$
 = $(474 - 324) \cdot 14 \cdot 1$
 = 21.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.4767 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.1824 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.1824 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[11.3, max(7.18, 4.5)]
 = 7.1824 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.48, 7.18)
 = 7.1824 mm

Available Nozzle Neck Thickness = $0.875 \cdot 14.275 = 12.490$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.2,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	268.8 N./mm ²	Passed
Occasional	:	7.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	15.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 14, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1080.00 / 5100.00 = 0.212$

Weld Size Calculations, Description: N3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	11.2748 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$7.0000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (12.7 - 32.3 + 2 \cdot 11.3 \cdot 0.86 \cdot (1 \cdot 15 - 4.18)) \cdot 137)$
 $= \max(0, -24338.24) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (6.33 + 21 + 2.3 - 0 \cdot 0.86) \cdot 137$
 $= 41393.16 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$



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$$= (6.33 + 0 + 0.86 + (2.91)) \cdot 137$$

$$= 14114.98 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (6.33 + 0 + 2.3 + 21 + (2.91)) \cdot 137$$

$$= 45462.98 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 474 \cdot 12 \cdot 0.49 \cdot 137$$

$$= 61140. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 156) \cdot (14.3 - 3) \cdot 0.7 \cdot 118$$

$$= 46626. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 324 \cdot 14 \cdot 0.74 \cdot 137$$

$$= 73622. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 324 \cdot (14 - 0) \cdot 0.74 \cdot 137$$

$$= 73623. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (61140 + 46626) = 107766 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (29992 + 73622 + 73623 + 0) = 177238 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (61140 + 73623 + 0) = 134763 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 107765 Kg, must exceed W = 0 Kg	or W1 = 41393 Kg
Path 2-2 = 177237 Kg, must exceed W = 0 Kg	or W2 = 14114 Kg
Path 3-3 = 134763 Kg, must exceed W = 0 Kg	or W3 = 45462 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2808 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.5950 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.5951 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	323.850	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	473.850	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1080.00	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{542.5(18.0 - 3.0)}$
 $= 621.537 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1085.000	mm
Nozzle Outside Diameter	(d) =	323.850	mm
Vessel Thickness used	(T) =	29.000	mm
Nozzle Thickness used	(t) =	11.275	mm
T / t	=	2.572	
d / t	=	28.723	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.826$$

Nr/P	=	0.083
Mr/P	=	0.079



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MØ/P = 0.061
 NØ/P = 0.181
 MrD/Mc = 0.169
 NrDL/Mc = 0.112
 MØd/Mc = 0.152
 NØDL/Mc = 0.183
 MrD/Ml = 0.105
 NrDL/Ml = 0.076
 MØD/Ml = 0.092
 NØDL/Ml = 0.309

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	-5	26	73	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-8	24	57	-26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	11	30	73	31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4



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Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-26	38	74	-29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-46	46	46	-46
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20
Inplane Stress Summary	-18	51	86	-32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	27	52	87	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17
Hoop Stress Summary	7	8	35	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	0	0	0	0



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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-122	122	122	-122
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-88	78	187	-136
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9

Shear Stress Summary	12	12	7	7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	98	80	188	174
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	13	14	30	31
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-208	208	208	-208
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-173	164	273	-222
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9

Shear Stress Summary	7	7	12	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	187	164	274	255

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{535.5(18.0 - 3.0)}$$

$$= 619.610 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1071.000 mm
Nozzle Outside Diameter	(d) =	473.850 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	86.275 mm
T / t	=	0.174
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.739$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.052
Mr/P	=	0.088
M0/P	=	0.026
N0/P	=	0.050
MrD/Mc	=	0.257
NrDL/Mc	=	0.094
M0d/Mc	=	0.077
N0DL/Mc	=	0.079
MrD/Ml	=	0.069
NrDL/Ml	=	0.055
M0D/Ml	=	0.021
N0DL/Ml	=	0.055

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	37	38	37	38
Outplane Stress Summary	37	35	79	19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	18	18	18	18



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Inplane Stress Summary	21	13	131	-71
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	38	36	132	90
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18

Outplane Stress Summary	-24	53	103	-37
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-182	182	182	-182
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	37	38	37	38

Inplane Stress Summary	-97	162	288	-180
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	98	162	288	180

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-10	-10	18	18	-3	-3	7	7
Circ. Q (SUS)	-16	16	34	-34	-34	34	58	-58
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-1	-1	5	5	-4	-4	12	12
Long. Q (SUS)	-17	17	41	-41	-33	33	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	9.6	10.4	37.0	38.0	16.1	17.1	26.2	27.2
Pm+Pl+Q (Total)	12.1	29.5	71.2	31.3	28.4	51.3	85.2	34.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	137.00	Passed
Pm+Pl (SUS)	38.04	205.50	Passed
Pm+Pl+Q (TOTAL)	85.21	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	37	38	37	38	37	38	37	38
Circ. Pl (SUS)	-2	-2	10	10	-6	-6	16	16
Circ. Q (SUS)	1	-1	29	-29	-130	130	234	-234
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-1	-1	11	11	-5	-5	13	13
Long. Q (SUS)	4	-4	100	-100	-39	39	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	37.0	38.0	37.0	38.0	37.0	38.0	37.0	38.0
Pm+Pl (SUS)	35.1	36.1	47.1	48.1	31.1	32.1	53.0	54.0



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Pm+Pl+Q (Total)	37.0	35.7	129.1	90.1	99.1	162.0	287.1	180.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	137.00	Passed
Pm+Pl (SUS)	54.04	205.50	Passed
Pm+Pl+Q (TOTAL)	287.09	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-10	-10	18	18	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-13	-13	17	17	-13	-13	17	17
Long. Q (SUS)	-83	83	161	-161	-169	169	247	-247
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	11.7	12.6	34.4	35.4	17.5	18.4	29.0	29.8
Pm+Pl+Q (Total)	96.5	79.7	186.3	171.6	186.5	164.3	272.5	252.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	35.39	177.00	Passed
Pm+Pl+Q (TOTAL)	272.49	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4-Shell Inlet

From: 50

Pressure for Reinforcement Calculations	P	1080.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1050.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6824.15	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

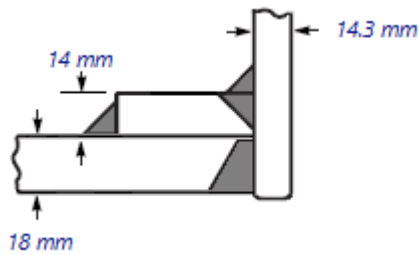
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	473.8500	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N4-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	0.562 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1080 \cdot 528 / (137 \cdot 1 - 0.6 \cdot 1080)$
 $= 4.1824 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1080 \cdot 162 / (118 \cdot 1 + 0.4 \cdot 1080)$
 $= 1.4767 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	602.6008 mm
Parallel to Vessel Wall, opening length	d	301.3004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.861, 1)$
 $= 0.861$



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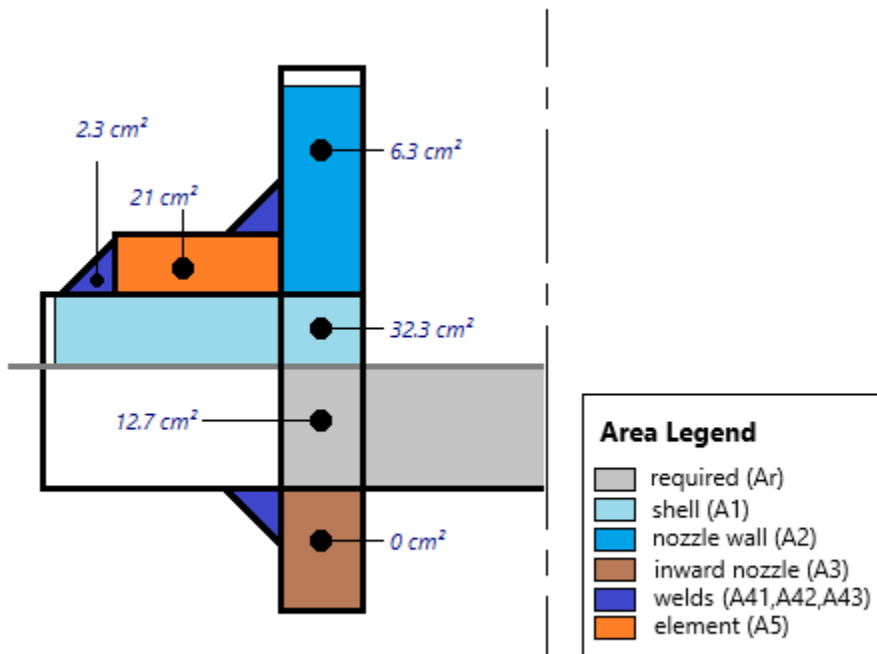


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.732	NA	NA
Area in Shell	A1	32.255	NA	NA
Area in Nozzle Wall	A2	6.329	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.301	NA	NA
Area in Element	A5	21.000	NA	NA
TOTAL AREA AVAILABLE	Atot	61.886	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (301 \cdot 4.18 \cdot 1 + 2 \cdot 11.3 \cdot 4.18 \cdot 1 \cdot (1 - 0.86))$$

$$= 12.732 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 301(1 \cdot 15 - 1 \cdot 4.18) - 2 \cdot 11.3$$

$$(1 \cdot 15 - 1 \cdot 4.18) \cdot (1 - 0.86)$$

$$= 32.255 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 37.5) \cdot (11.3 - 1.48) \cdot 0.86$$



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= 6.329 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$
 = 2.301 cm²

Area Available in Element [A5]:
 = $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$
 = $(474 - 324) \cdot 14 \cdot 1$
 = 21.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.4767 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.1824 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.1824 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[11.3, max(7.18, 4.5)]
 = 7.1824 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.48, 7.18)
 = 7.1824 mm

Available Nozzle Neck Thickness = 0.875 * 14.275 = 12.490 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.2,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	268.8 N./mm ²	Passed
Occasional	:	7.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	15.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 14, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.28$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 12.5, tr = 1.48, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1080.00 / 5100.00 = 0.212$

Weld Size Calculations, Description: N4-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	11.2748 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$7.0000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (12.7 - 32.3 + 2 \cdot 11.3 \cdot 0.86 \cdot (1 \cdot 15 - 4.18)) \cdot 137)$
 $= \max(0, -24338.24) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (6.33 + 21 + 2.3 - 0 \cdot 0.86) \cdot 137$
 $= 41393.16 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$



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$$= (6.33 + 0 + 0.86 + (2.91)) \cdot 137$$

$$= 14114.98 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (6.33 + 0 + 2.3 + 21 + (2.91)) \cdot 137$$

$$= 45462.98 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 474 \cdot 12 \cdot 0.49 \cdot 137$$

$$= 61140. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 156) \cdot (14.3 - 3) \cdot 0.7 \cdot 118$$

$$= 46626. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 324 \cdot 14 \cdot 0.74 \cdot 137$$

$$= 73622. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 324 \cdot (14 - 0) \cdot 0.74 \cdot 137$$

$$= 73623. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (61140 + 46626) = 107766 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (29992 + 73622 + 73623 + 0) = 177238 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (61140 + 73623 + 0) = 134763 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 107765 Kg, must exceed W = 0 Kg	or W1 = 41393 Kg
Path 2-2 = 177237 Kg, must exceed W = 0 Kg	or W2 = 14114 Kg
Path 3-3 = 134763 Kg, must exceed W = 0 Kg	or W3 = 45462 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2808 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 25.5950 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.5951 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Inlet:



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	323.850	mm
Nozzle Thickness	14.2748	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	473.850	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1080.00	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{542.5(18.0 - 3.0)}$
 $= 621.537 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1085.000	mm
Nozzle Outside Diameter	(d) =	323.850	mm
Vessel Thickness used	(T) =	29.000	mm
Nozzle Thickness used	(t) =	11.275	mm
T / t	=	2.572	
d / t	=	28.723	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.826$$

Nr/P	=	0.083
Mr/P	=	0.079



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MØ/P = 0.061
 NØ/P = 0.181
 MrD/Mc = 0.169
 NrDL/Mc = 0.112
 MØd/Mc = 0.152
 NØDL/Mc = 0.183
 MrD/Ml = 0.105
 NrDL/Ml = 0.076
 MØD/Ml = 0.092
 NØDL/Ml = 0.309

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	19	20	19	20
Outplane Stress Summary	-5	26	73	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-8	24	57	-26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	11	30	73	31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	4	4	4	4



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Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-26	38	74	-29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-46	46	46	-46
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	19	20	19	20
Inplane Stress Summary	-18	51	86	-32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	27	52	87	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17
Hoop Stress Summary	7	8	35	37

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	0	0	0	0



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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-122	122	122	-122
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-88	78	187	-136
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	9	9	9	9
Shear Stress Summary	12	12	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	98	80	188	174

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17
Hoop Stress Summary	13	14	30	31

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-208	208	208	-208
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-173	164	273	-222

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	9	9	9	9
Shear Stress Summary	7	7	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl



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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	187	164	274	255

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{535.5(18.0 - 3.0)}$$

$$= 619.610 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1071.000 mm
Nozzle Outside Diameter	(d) =	473.850 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	86.275 mm
T / t	=	0.174
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.739$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.052
Mr/P	=	0.088
M0/P	=	0.026
N0/P	=	0.050
MrD/Mc	=	0.257
NrDL/Mc	=	0.094
M0d/Mc	=	0.077
N0DL/Mc	=	0.079
MrD/Ml	=	0.069
NrDL/Ml	=	0.055
M0D/Ml	=	0.021
N0DL/Ml	=	0.055

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	37	38	37	38
Outplane Stress Summary	37	35	79	19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	18	18	18	18



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Inplane Stress Summary	21	13	131	-71
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	38	36	132	90
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-54	54	54	-54
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18

Outplane Stress Summary	-24	53	103	-37
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-182	182	182	-182
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	37	38	37	38

Inplane Stress Summary	-97	162	288	-180
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	98	162	288	180

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-10	-10	18	18	-3	-3	7	7
Circ. Q (SUS)	-16	16	34	-34	-34	34	58	-58
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-1	-1	5	5	-4	-4	12	12
Long. Q (SUS)	-17	17	41	-41	-33	33	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	9.6	10.4	37.0	38.0	16.1	17.1	26.2	27.2
Pm+Pl+Q (Total)	12.1	29.5	71.2	31.3	28.4	51.3	85.2	34.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	137.00	Passed
Pm+Pl (SUS)	38.04	205.50	Passed
Pm+Pl+Q (TOTAL)	85.21	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	37	38	37	38	37	38	37	38
Circ. Pl (SUS)	-2	-2	10	10	-6	-6	16	16
Circ. Q (SUS)	1	-1	29	-29	-130	130	234	-234
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-1	-1	11	11	-5	-5	13	13
Long. Q (SUS)	4	-4	100	-100	-39	39	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	37.0	38.0	37.0	38.0	37.0	38.0	37.0	38.0
Pm+Pl (SUS)	35.1	36.1	47.1	48.1	31.1	32.1	53.0	54.0



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Pm+Pl+Q (Total)	37.0	35.7	129.1	90.1	99.1	162.0	287.1	180.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	137.00	Passed
Pm+Pl (SUS)	54.04	205.50	Passed
Pm+Pl+Q (TOTAL)	287.09	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-10	-10	18	18	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-13	-13	17	17	-13	-13	17	17
Long. Q (SUS)	-83	83	161	-161	-169	169	247	-247
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	11.7	12.6	34.4	35.4	17.5	18.4	29.0	29.8
Pm+Pl+Q (Total)	96.5	79.7	186.3	171.6	186.5	164.3	272.5	252.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	35.39	177.00	Passed
Pm+Pl+Q (TOTAL)	272.49	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N5-Shell Outlet

From: 50

Pressure for Reinforcement Calculations	P	1090.292	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1050.00	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4924.15	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

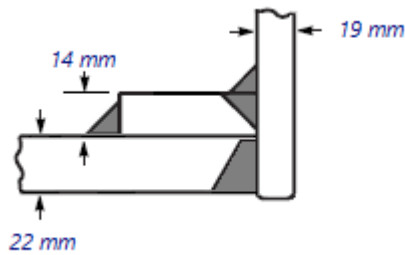
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	607.2000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: N5-Shell Outlet Nozl: 15 9:43am Sep 9,2026



Reinforcement CALCULATION, Description: N5-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1090 \cdot 528 / (137 \cdot 1 - 0.6 \cdot 1090)$
 $= 4.2224 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1090 \cdot 229 / (118 \cdot 1 + 0.4 \cdot 1090)$
 $= 2.1046 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	850.2000 mm
Parallel to Vessel Wall, opening length	d	425.1000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 137)$
 $= 0.861$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137 / 137)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.861$



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Nozzle Calcs.: N5-Shell Outlet

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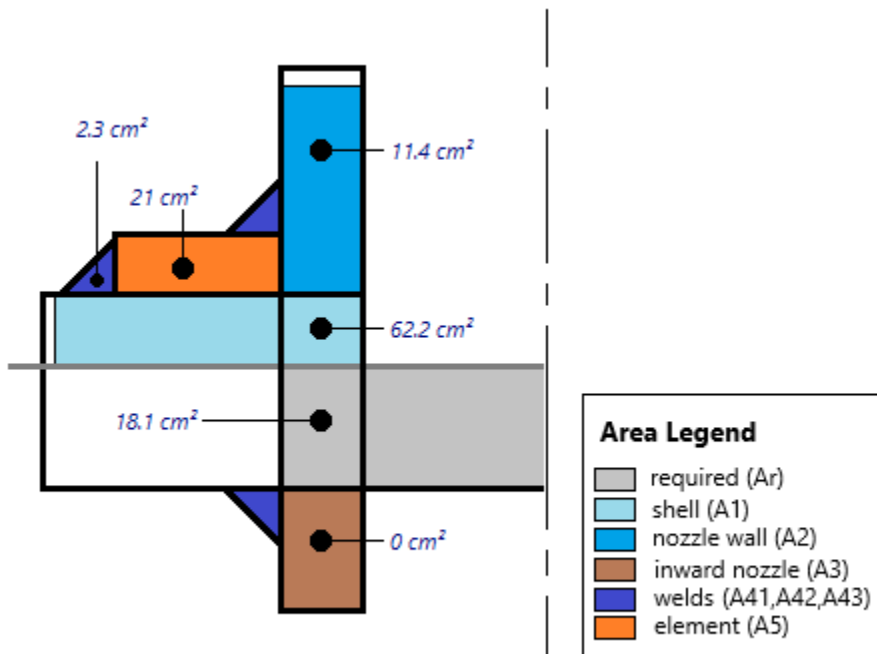


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.137	NA	NA
Area in Shell	A1	62.162	NA	NA
Area in Nozzle Wall	A2	11.411	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.301	NA	NA
Area in Element	A5	21.000	NA	NA
TOTAL AREA AVAILABLE	Atot	96.873	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (425 \cdot 4.22 \cdot 1 + 2 \cdot 16 \cdot 4.22 \cdot 1 \cdot (1 - 0.86)) \\
 &= 18.137 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 425(1 \cdot 19 - 1 \cdot 4.22) - 2 \cdot 16 \\
 &\quad (1 \cdot 19 - 1 \cdot 4.22) \cdot (1 - 0.86) \\
 &= 62.162 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 47.5) \cdot (16 - 2.1) \cdot 0.86
 \end{aligned}$$



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$$= 11.411 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$
 $= 2.301 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 $= (607 - 457) \cdot 14 \cdot 1$
 $= 21.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.1046 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 7.2224 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 7.2224 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[11.3, \max(7.22, 4.5)]$
 $= 7.2224 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(5.1, 7.22)$
 $= 7.2224 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 19.050 = 16.669 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	22.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	272.6 N./mm ²	Passed
Occasional	:	7.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	11.9,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 16.7$, $t_r = 2.1$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.15$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	7°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 14$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.22$, Temp. Reduction = 78°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C
Min Metal Temp. w/o impact per UG-20(f)	-29°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, tg = 14, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.22$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:D)

Govrn. thk, tg = 14, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.22$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 16.7, tr = 2.1, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1090.29 / 5100.00 = 0.214$

Weld Size Calculations, Description: N5-Shell Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	14.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	14.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * Wo \text{ mm}$
Pad Weld	$7.0000 = 0.5 * TminPad$	$8.4840 = 0.7 * Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)$
 $= \max(0, (18.1 - 62.2 + 2 * 16 * 0.86 * (1 * 19 - 4.22))137)$
 $= \max(0, -55793.46) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2*fr2)*Sv$
 $= (11.4 + 21 + 2.3 - 0 * 0.86) * 137$
 $= 48491.66 \text{ Kgf}$

Weld Load [W2]:



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$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (11.4 + 0 + 0.86 + (5.25)) \cdot 137$$

$$= 24482.11 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (11.4 + 0 + 2.3 + 21 + (5.25)) \cdot 137$$

$$= 55830.11 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 457 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 42342. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 607 \cdot 12 \cdot 0.49 \cdot 137$$

$$= 78346. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 221) \cdot (19 - 3) \cdot 0.7 \cdot 118$$

$$= 93675. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 457 \cdot 14 \cdot 0.74 \cdot 137$$

$$= 103938. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 457 \cdot (14 - 0) \cdot 0.74 \cdot 137$$

$$= 103939. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (78346 + 93675) = 172021 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (42342 + 103938 + 103939 + 0) = 250218 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (78346 + 103939 + 0) = 182285 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 172021 Kgf, must exceed W = 0 Kgf	or W1 = 48491 Kgf
Path 2-2 = 250218 Kgf, must exceed W = 0 Kgf	or W2 = 24482 Kgf
Path 3-3 = 182284 Kgf, must exceed W = 0 Kgf	or W3 = 55830 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2808 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 52.3828 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 274.3828 mm

Input Echo, WRC297 Item 1, Description: N5-Shell Outle:



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	457.200	mm
Nozzle Thickness	19.0500	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	607.200	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1090.29	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{544.5(22.0 - 3.0)}$
 $= 792.852 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1089.000	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	16.050	mm
T / t	=	2.056	
d / t	=	28.486	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.412$$

$$Nr/P = 0.086$$



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Mr/P = 0.079
 MØ/P = 0.048
 NØ/P = 0.151
 MrD/Mc = 0.182
 NrDL/Mc = 0.112
 MØd/Mc = 0.125
 NØDL/Mc = 0.180
 MrD/Ml = 0.104
 NrDL/Ml = 0.067
 MØD/Ml = 0.063
 NØDL/Ml = 0.274

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	16	18	16	18
Outplane Stress Summary	-6	19	67	7

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-35	35	35	-35
Inplane Pressure Stress	8	8	8	8
Inplane Stress Summary	-13	28	65	-35

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	15	30	69	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	8	8	8	8
Outplane Stress Summary	-30	36	74	-28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-62	62	62	-62
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	16	18	16	18
Inplane Stress Summary	-34	61	102	-49

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	35	61	103	50

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	18	16	18
Hoop Stress Summary	5	7	36	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	28	-28	28	-28



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Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-102	102	102	-102
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-76	72	152	-108

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	9	9	5	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	84	73	152	147

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	18	16	18
Hoop Stress Summary	11	12	31	32

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	-11	-11	11	11
Axial Bending (Mc)	-181	181	181	-181
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-155	151	231	-188

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	7	7	7	7
Shear Stress Summary	5	5	9	9

Nozzle Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	167	151	232	221

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{537.5(22.0 - 3.0)}$
 $= 790.688 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1075.000 mm
 Nozzle Outside Diameter (d) = 607.200 mm
 Vessel Thickness used (T) = 19.000 mm
 Nozzle Thickness used (t) = 91.050 mm
 T / t = 0.209
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.249$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.047
 Mr/P = 0.077
 M0/P = 0.025
 N0/P = 0.052
 MrD/Mc = 0.252
 NrDL/Mc = 0.088
 M0d/Mc = 0.076
 N0DL/Mc = 0.083
 MrD/Ml = 0.065
 NrDL/Ml = 0.044
 M0D/Ml = 0.020
 N0DL/Ml = 0.053

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	29	30	29	30
Outplane Stress Summary	26	30	70	13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	43	-43	43	-43
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-50	50	50	-50



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Inplane Pressure Stress	14	14	14	14
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Inplane Stress Summary	6	20	118	-68
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	27	32	118	82
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14

Outplane Stress Summary	-35	53	102	-41
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	43	-43	43	-43
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-195	195	195	-195
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	29	30	29	30

Inplane Stress Summary	-129	176	284	-192
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	129	176	284	192

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	18	16	18	16	18	16	18
Circ. Pl (SUS)	-11	-11	19	19	-4	-4	8	8
Circ. Q (SUS)	-13	13	29	-29	-48	48	76	-76
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-1	-1	5	5	-6	-6	14	14
Long. Q (SUS)	-21	21	49	-49	-34	34	50	-50
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	16.0	18.0	16.0	18.0	16.0	18.0	16.0	18.0
Pm+Pl (SUS)	7.0	7.0	35.0	37.0	12.0	14.0	24.0	26.0
Pm+Pl+Q (Total)	15.2	29.0	66.2	44.4	37.6	62.3	100.3	50.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	137.00	Passed
Pm+Pl (SUS)	37.00	205.50	Passed
Pm+Pl+Q (TOTAL)	100.32	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	-2	-2	10	10	-7	-7	15	15
Circ. Q (SUS)	-2	2	28	-28	-152	152	238	-238
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-1	-1	9	9	-6	-6	14	14
Long. Q (SUS)	-7	7	93	-93	-45	45	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	27.1	28.1	39.1	40.1	22.1	23.1	44.1	45.1



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Pm+Pl+Q (Total)	25.8	31.4	116.1	82.1	130.0	175.0	282.1	193.1
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	137.00	Passed
Pm+Pl (SUS)	45.06	205.50	Passed
Pm+Pl+Q (TOTAL)	282.09	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	18	16	18	16	18	16	18
Circ. Pl (SUS)	-11	-11	19	19	-6	-6	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-10	-10	12	12	-10	-10	12	12
Long. Q (SUS)	-74	74	130	-130	-153	153	209	-209
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	16.0	18.0	16.0	18.0	16.0	18.0	16.0	18.0
Pm+Pl (SUS)	7.3	9.2	35.1	37.1	12.2	14.1	30.1	32.1
Pm+Pl+Q (Total)	82.6	73.0	150.3	147.5	165.4	151.3	229.3	221.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	118.00	Passed
Pm+Pl (SUS)	37.06	177.00	Passed
Pm+Pl+Q (TOTAL)	229.32	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 ASME TS Calc: DESIGN CASE Case: 1 9:43a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	1080.00 KPa.
Shell Maximum Operating Pressure	Psox,max	420.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1050.000 mm
Shell Temperature for Internal Pressure	Ts	275.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	540.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1050.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	668
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	7350.00 mm
Straight Tube Length (bet. inner tubsht faces) L		7263.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		74.15 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	111.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	35.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	505.621 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	7204.8 cm ²
Perimeter of Tube Layout	Cp	3196.3 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	132.50 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	87.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1230.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	1226.00 mm
Step 1 Depth on the Tubeside	ht1	0.00 mm
Step 2 Diameter on the Tubeside	dt2	1226.00 mm
Step 2 Depth on the Tubeside	ht2	0.00 mm
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	1226.00 mm
Step 1 Depth on the Shellside	hs1	0.00 mm
Step 2 Diameter on the Shellside	ds2	1226.00 mm
Step 2 Depth on the Shellside	hs2	0.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		1230.00 mm

Note: Tubesheet diameter is now: 1230.000 mm Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	1099.745 mm
Area of the Untubed Lanes	AL	526.9 cm ²

Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	103.0000 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	1230.000 mm
Flange Inside Diameter	B	1050.000 mm
Flange Face Outside Diameter	Fod	1125.000 mm
Flange Face Inside Diameter	Fid	1050.000 mm
Gasket Outside Diameter	Go	1114.000 mm
Gasket Inside Diameter	Gi	1082.000 mm
Small end Hub thk.	g0	14.0000 mm
Large end Hub thk.	g1	20.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	1102.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²



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Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

Bolting Information:

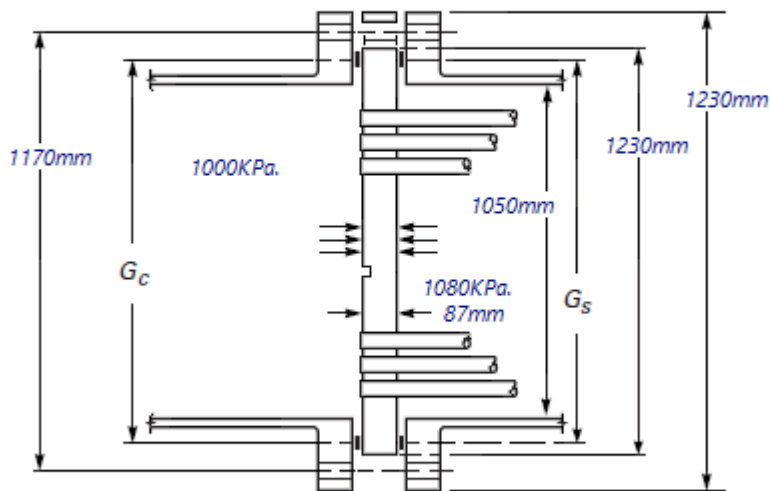
Diameter of Bolt Circle	C	1170.000 mm
Nominal Bolt Diameter	dB	28.5750 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	44
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	129912.99 Kg
Alternate Flange Seating Bolt Load, Wm2	215021.02 Kg
Alternate Flange Design Bolt Load, W	362446.06 Kg

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 55.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.



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Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.17745E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	275.0 °C	0.18674E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000 \cdot (25.4/2 - 0)) / (74.2 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.1704 + 0.0000 = 0.1704 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	275.00 °C
Elastic Modulus for Material			172442880.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7768.62	15.39	50.0000	0.0046419	45.13

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 45.1) / (3 \cdot 15.4) = 3909 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6813	25.40	7768.62	37.28	50.0000	0.0007913	33.09

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 33.1) / (3 \cdot 37.3) = 1183 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6813 mm
Allowable Internal Pressure at Corroded thickness	10161.22 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	3908.62 KPa.
Required External Design Pressure	1183.00 KPa.
Required Thickness due to Shell Side pressure	0.6813 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	16.000 mm
Basic Gasket Width,	$b0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	1099.750 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 506 + 25.4$
 $= 1036.642 \text{ mm}$



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Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (35 - 25.4) / 35$$

$$= 0.274$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1100 / 1037$$

$$= 1.0609$$

Ratio [ρ_c]:

$$= G_c / D_o \text{ (Configurations d)}$$

$$= 1100 / 1037$$

$$= 1.0609$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 87$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t_t (E_t / E) \cdot (S_t / S) \cdot (\rho); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 1.65 (177454944 / 186735680) \cdot (74.2 / 133) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 35 / \sqrt{ 1 - 4 \cdot \min(527; 4 \cdot 1037 \cdot 35) / (\pi \cdot 1037^2) }$$

$$= 36.1465 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (36.1 - 25.4) / 36.1$$

$$= 0.2973$$

$$h/p = 2.485714 ; \mu^* = 0.297304 ; E^* = 51841608 \text{ KPa.}$$

$$E^*/E = 0.277620 ; v^* = 0.355697$$

Note: As h/p (2.486) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1230 / 1037$$

$$= 1.1865$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.36) / 51841608 \cdot (186735680 \cdot \ln(1.19))$$

$$= 0.3969$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1037^2 / 16 \cdot [(1.06 - 1) \cdot (1.06^2 + 1) \cdot 1080 -$$



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$$(1.06 - 1) \cdot (1.06^2 + 1) \cdot 1000]$$

$$= 696005.1250 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]
 $= Mts + W* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$
 $= 696005 + 362446 \cdot (1100 - 1100) / (2 \cdot \pi \cdot 1037)$
 $= 696005.1250 \text{ KPa} \cdot \text{mm}^2$

Note: W is the maximum of the bolt loads between the shell and channel sides.*

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= (M* - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= (696005) - 1037^2/32 \cdot 0.4 (1080 - 1000) / (1 + 0.4)$
 $= -265119.3750 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$
 $= -265119 + 1037^2/64 \cdot (3 + 0.36) \cdot (1080 - 1000)$
 $= 4242534.5000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(|Mp|; |Mo|)$
 $= \max(|-265119|; |4242534|)$
 $= 4242534.5000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 4242534 / (0.3 (87 - 8)^2)$
 $= 13.7198 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S$
 $= 2 \cdot 133$
 $= 265.00 \text{ N./mm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 4183329 / (0.3 (25.9 - 8)^2)$
 $= 264.3762 \text{ N./mm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 25.9 + 0 + 0$
 $= 25.8705 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(25.9; 0.62)$
 $= 25.8705 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|Ps - Pt|$]:
 $= |Ps - Pt|$
 $= |1080 - 1000|$
 $= 80.000 \text{ KPa.}$

Shear Stress check [τ_{limit}]:
 $= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$
 $= 4 \cdot 0.27 \cdot 87/1037 \cdot \min(0.8 \cdot 133; 0.533 \cdot 199)$
 $= 9.760 \text{ N./mm}^2$



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| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.27 \cdot 106) \cdot (4 \cdot 7205 / 3196) \cdot | 1080 - 1000 | + 0$$

$$= 0.6203 \text{ mm}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 74.2$$

$$= 930.855 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 74.2$$

$$= 883.0262 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 74.2 / 74.2$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 74.2$$

$$= 571.3698 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ ((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd) } - 0.75 \cdot do)$$

$$= 2 (\sqrt{ ((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1) } - 0.75 \cdot 25.4)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
 Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 931 is ≥ tube strength 931 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	13.7 ≤	265.0 N./mm ²	Ok
Tubesheet shear stress	0.8 ≤	106.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
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Tubesheet Thickness :	25.871	87.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	71.551 ...	171	265	9 106	Ps+Pt D1	Ok
D2uc	77.236 19.768	203	265	11 106	Ps+Pt D2	Ok
D3uc	25.871 ...	14	265	1 106	Ps+Pt D3	Ok
D4uc	28.295 ...	18	265	1 106	Ps+Pt D4	Ok
D1c	74.614 ...	185	265	10 106	Ps+Pt-c D1	Ok
D2c	80.152 19.768	219	265	12 106	Ps+Pt-c D2	Ok
D3c	28.815 ...	15	265	1 106	Ps+Pt-c D3	Ok
D4c	31.292 ...	19	265	1 106	Ps+Pt-c D4	Ok
Max:	80.1516 19.768 mm	0.827	0.113	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	80.152	87.000	Ok
Tubesheet thickness flanged extension :	19.768	55.000	Ok
Tube thickness :	0.681	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	1430.28	1.000	1430.27	1.000
Tubesheet Shear Stress	10447.26	1.000	10447.26	1.000
Tube Pressure Stress	10161.21	1.000	3908.61	1.000
Tubesheet Extension Stress	No Calc	No Calc	3143.51	...
Minimum MAWP	1430.28		1430.27	



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Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1609.54	1.000	1609.54	1.000
Tubesheet Shear Stress	11686.63	1.000	11686.63	1.000
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Tubesheet Extension Stress	No Calc	No Calc	3143.51	...
Minimum MAPnc	1609.54		1609.54	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
 = tubesheet thickness/4
 = 87/4
 = 21.750 mm

Thickness Ratio = 0.67, Temperature Reduction per Fig. UCS 66.1 = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(1000/1610, 1080/1610)
 = 0.671

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: